CHAPTER 12

The Time-Dependent Hartree–Fock Method (TDHF)

12.1 Introduction

On the way to describe processes in nuclei involving large amplitudes like anharmonic vibrations or fission and fusion processes, up to now we have only used stationary wave functions together with the Ritz variational principle.

On the other side, we can also formulate the problem in a time-dependent way. Let us start at time $t=0$ with a wave function $|\Psi(0)\rangle$ which is not an exact solution of the stationary many-body Schrödinger equation. Such a state represents a *wave packet*, and its behavior with increasing time is given by the time evolution operator:

$$|\Psi(t)\rangle = e^{-iHt/\hbar}|\Psi(0)\rangle, \tag{12.1}$$

where H is the full many-body Hamiltonian. The information contained in the time-dependent wave function $|\Psi(t)\rangle$ is, in principle, equivalent to the solution of the stationary Schrödinger equation, since a Fourier analysis of $|\Psi(t)\rangle$ gives us the spectrum of H and the corresponding eigenstates. On the other hand, the description of the system by a wave packet is much closer to our classical intuition. In the fission process, for instance, we have a classical droplet in mind which at each time has a fixed deformation. In a microscopic picture, such a deformed state would be represented by a Slater determinant in a deformed potential well, which is certainly no eigenstate of H, but a wave packet in the sense of Eq. (12.1). In fact, we

will see that in such a picture, under certain approximations, classical equations of motion can be derived.

The general ansatz (12.1) is certainly much too complicated for practical solutions. Therefore, in this chapter, we will make the following clear-cut approximations in two successive steps:

(i) Within the *time-dependent Hartree–Fock* (TDHF) theory we start with a Slater determinant $|\Phi(0)\rangle$ and assume that it stays a Slater determinant for all times. In general, this implies a drastic approximation and yields, as we shall see, serious problems in the interpretation of the results. On the other hand, in the static case we have seen that this assumption is based on the fact that because of the Pauli principle each particle does not interact with the other particles directly, but only with the wall of a self-consistent field. Together with the Brückner renormalization of the force and the inclusion of pairing correlation by HFB theory, this approximation yields an excellent description of ground states and single-particle energies throughout the periodic table. At least for energies per particle well below the Fermi energy, the Pauli principle should work in a similar way in the time-dependent case, and there is no reason why the HF approximation should break down in the dynamic case. There is a wide range of phenomena, such as low-lying collective excitations, nuclear fission, and heavy ion collisions at moderate energies, for which this approach seems reasonable. Therefore, we shall in Section 12.2 derive the TDHF equations, which are the basis of this theory, and discuss their properties and the problems involved. We will also present several applications. Naturally, density-dependent forces are used, and in many cases pairing correlations also have to be included [Be 66, BF 76]. Since it will become obvious how to do such calculations, we will present the whole theory for Slater determinants without pairing ($\rho^2=\rho$) only.

(ii) In a second step, we use the adiabatic approximation, that is, we assume that the velocities in our systems are small. An expansion in these velocities up to second order yields the *adiabatic time-dependent Hartree–Fock* (ATDHF) theory. It results in equations that, in principle, are no easier to solve than the TDHF equations, but it gives much insight into the physical problems and allows the derivation of a collective Hamiltonian for all cases in which the motion may be described in terms of a few collective variables. This will be discussed in Section 12.3.

12.2 The Full Time-Dependent Hartree–Fock Theory

12.2.1 Derivation of the TDHF Equation

We start with an arbitrary wave packet $|\Psi(0)\rangle$. Its exact time evolution is given by Eq. (12.1) and its one-body density at time t is given by

$$\rho_{kl}(t)=\langle\Psi(t)|c_l^+ c_k|\Psi(t)\rangle. \tag{12.2}$$

To obtain an equation of motion, we calculate its time derivative:

$$i\hbar\dot{\rho}_{kl}(t)=\langle\Psi(t)|\left[c_l^+c_k, H\right]|\Psi(t)\rangle. \tag{12.3}$$

For a two-body Hamiltonian [Eq. (5.25)] it can be written as

$$i\hbar\dot{\rho}_{kl}-\sum_p t_{kp}\rho_{pl}-\rho_{kp}t_{pl}=\frac{1}{2}\sum_{prs}\bar{v}_{kprs}\rho^{(2)}_{rslp}-\bar{v}_{rslp}\rho^{(2)}_{kprs}, \tag{12.4}$$

where we have introduced the two-body density

$$\rho^{(2)}_{klpq}(t)=\langle\Psi(t)|c_p^+c_q^+c_lc_k|\Psi(t)\rangle. \tag{12.5}$$

We can derive an equation of motion for this quantity in the same way as we did for the one-body density in Eq. (12.3). It connects the two-body density with the three-body density, and so on (see Appendix F). This system of equations [BG 49] is exact, but certainly not closed. In order to close it, we have to introduce some approximations. The l.h.s. of Eq. (12.4) describes the free motion of the system. It contains only the kinetic energy, whereas the r.h.s. contains the interaction between the particles. This interaction can be divided into two parts— a self-consistent one-body field, which averages over all particles, and individual collisions between two nucleons which cannot be taken into account by the mean field. The latter causes the two-body correlations.

To achieve this decomposition we define a correlation function $g^{(2)}$ by extracting the uncorrelated pairs of $\rho^{(2)}$,

$$\rho^{(2)}_{klpq}=\rho_{kp}\rho_{lq}-\rho_{kq}\rho_{lp}+g^{(2)}_{klpq}, \tag{12.6}$$

and gain, instead of Eq. (12.4),

$$i\hbar\dot{\rho}_{kl}-\left[t+\Gamma,\rho\right]_{kl}=\frac{1}{2}\sum_{prs}\bar{v}_{kprs}g^{(2)}_{rslp}-\bar{v}_{rslp}g^{(2)}_{kprs}, \tag{12.7}$$

with the density dependent HF potential

$$\Gamma_{kl}=\sum_{pq}\bar{v}_{kqlp}\rho_{pq}, \tag{12.8}$$

which is defined as in the static HF theory [Eq. (5.34)]. The only difference is that it now contains the exact time-dependent density $\rho(t)$; therefore, the self-consistent field Γ is time dependent.

Particles in this system can now undergo two kinds of interactions— collisions with the moving walls of the self-consistent field Γ or collisions with the other particles. Both types provide internal excitations of the system. We therefore have to distinguish between "*one-body friction*," caused by the collisions with the wall (see [MSK 77]), and "*two-body friction*," caused by two-body collisions on the r.h.s. of Eq. (12.7).

As we discussed in the Introduction, for the static case the two-body correlations can be neglected because of the Pauli principle. They can be summed up in a properly renormalized effective interaction of a Brückner theory. We therefore also expect that two-body scattering plays no essential role in the dynamical case, as long as the excitation energies per particle are less than the Fermi energy and $g^{(2)}$ may be neglected. In the

classical limit ($\hbar \to 0$) this corresponds to the neglect of the collision term in the Boltzmann equation, as we will discuss in more detail in Chapter 13. In this case, the two-body density $\rho^{(2)}$ in Eq. (12.6) factorizes and can be expressed completely by the one-body density ρ; the wave function $|\Psi(t)\rangle$ then stays a Slater determinant $|\Phi(t)\rangle$ for all times. For $g^{(2)}=0$ we therefore end up with the *TDHF equation* [Di 30]

$$i\hbar\dot{\rho} = [h, \rho] \tag{12.9}$$

with $h = t + \Gamma$. It is a nonlinear matrix equation for the one-body density matrix ρ and a first-order differential equation in time. Starting with an initial condition $\rho(0)$, it defines $\rho(t)$ for all times.*

In the case of an external field $F(t) = \sum_{kl} f_{kl}(t) c_k^+ c_l$, we have the Schrödinger equation

$$i\hbar|\dot{\Psi}(t)\rangle = (H + F(t))|\Psi(t)\rangle \tag{12.10}$$

that is, we have to replace the Hamiltonian H in Eq. (12.3) by $H + F(t)$, or the matrix h in Eq. (12.9) by $h + f(t)$. This was used to derive linear response theory in a time-dependent external field (see Sec. 8.5).

In *coordinate space* (it is obvious how to also include spin and isospin variables) with a spin independent two-body potential $v(\mathbf{r}, \mathbf{r}')$ the TDHF equation (12.9) has the form

$$\begin{aligned} i\hbar\dot{\rho}(\mathbf{r}, \mathbf{r}', t) = {} & \frac{-\hbar^2}{2m}(\Delta_{\mathbf{r}} - \Delta_{\mathbf{r}'})\rho(\mathbf{r}, \mathbf{r}', t) + (\Gamma_H(\mathbf{r}) - \Gamma_H(\mathbf{r}')) \cdot \rho(\mathbf{r}, \mathbf{r}', t) \\ & + \int d^3r''(\Gamma_{Ex}(\mathbf{r}, \mathbf{r}'')\rho(\mathbf{r}'', \mathbf{r}', t) - \Gamma_{Ex}(\mathbf{r}'', \mathbf{r}')\rho(\mathbf{r}, \mathbf{r}'', t)); \end{aligned} \tag{12.11}$$

where Γ_H and Γ_{Ex} are defined as in Eq. (5.42)ff.

We see that we get a solution $\rho(t)$ of Eq. (12.9) if we decompose the Slater determinant $|\Phi(0)\rangle$ into A orthogonal single-particle wave functions $\varphi_i(\mathbf{r}, 0)$ $(i = 1, \ldots, A)$ and follow their time evolution by the TDHF equation for single-particle wave functions:

$$i\hbar\frac{\partial}{\partial t}\varphi_i(\mathbf{r}, t) = \left(-\frac{\hbar^2}{2m}\Delta + \Gamma_H(\mathbf{r}, t)\right)\varphi_i(\mathbf{r}, t) + \int d\mathbf{r}'\, \Gamma_{Ex}(\mathbf{r}, \mathbf{r}', t)\varphi_i(\mathbf{r}', t). \tag{12.12}$$

The density ρ is then given by

$$\rho(\mathbf{r}, \mathbf{r}', t) = \sum_{i=1}^{A} \varphi_i(\mathbf{r}, t) \cdot \varphi_i^*(\mathbf{r}', t). \tag{12.13}$$

This form of the TDHF equation is easier to handle for practical calculations, because it is simpler to solve A differential equations depending on four coordinates $(\mathbf{r}, t)$ than one differential equation depending on seven coordinates $(\mathbf{r}, \mathbf{r}', t)$ as in Eq. (12.11).

*This equation has been explicitly used in nuclear physics for the first time by Ferrel [Fe 57].

Before we discuss the properties of the TDHF equations, we should mention that it can also be derived from a time-dependent *variational principle* [KK 76]. The Lagrangian is defined as

$$\mathfrak{L} = \langle \Phi(t) | i\hbar \frac{\partial}{\partial t} - H | \Phi(t) \rangle. \tag{12.14}$$

Under the assumption that $|\Phi(t)\rangle$ is a Slater determinant* of the single-particle functions $\varphi_1(\mathbf{r}, t), \ldots, \varphi_A(\mathbf{r}, t)$, $\mathfrak{L}$ is a functional of the $2A$ independent fields $\varphi_i(\mathbf{r}, t)$ and $\varphi_i^*(\mathbf{r}, t)$. Using them as classical field variables, we can define an action

$$I_{12} = \int_{t_1}^{t_2} \mathfrak{L}[\varphi, \varphi^*]\, dt \tag{12.15}$$

with fixed endpoints 1 and 2. The Hamilton variational principle [Go 59]

$$\delta I_{12} = 0 \tag{12.16}$$

with independent variation of φ and φ^* gives the equations of motion (12.12).

12.2.2 Properties of the TDHF Equation

12.2.2.1. Conservation of Product Character. In the last section we assumed that the wave function $|\Phi(t)\rangle$ is a Slater determinant for all times and under this condition we derived the TDHF equations (12.9). Therefore, the first thing we have to show is that the theory is consistent, that is, if we start with a density of a Slater determinant ($\rho^2(0) = \rho(0)$) this property is conserved for all times:

$$i\hbar \frac{\partial}{\partial t}(\rho^2 - \rho) = i\hbar(\dot{\rho}\rho + \rho\dot{\rho} - \dot{\rho}) = [h, \rho^2 - \rho]. \tag{12.17}$$

In Chapter 7 we discussed in great detail the energy surface which we get if all the Slater determinants are parameterized in a suitable way. The solution of the TDHF equation $|\Phi(t)\rangle$ therefore corresponds to a *trajectory* on this energy surface.

12.2.2.2. Conservation of Orthogonality. Closely connected to the property just discussed is the fact that single-particle wave functions $\varphi_i(\mathbf{r}, t)$ stay orthogonal for all times:

$$i\hbar \frac{\partial}{\partial t} \langle \varphi_i | \varphi_k \rangle = \langle \varphi_i | h - h^+ | \varphi_k \rangle = 0. \tag{12.18}$$

12.2.2.3. Conservation of Expectation Values of Symmetry Operators. For an arbitrary, time-independent, single-particle operator $F = \sum f_{lm} c_l^+ c_m$,

*This variational principle can be generalized. If $|\Phi\rangle$ is a general many-body wave function, the exact Schrödinger equation [KK 76] can be derived.

from Eq. (12.9) we obtain for the expectation values in the state $|\Phi(t)\rangle$

$$i\hbar\frac{d}{dt}\langle F\rangle = i\hbar\,\mathrm{Tr}(f\cdot\dot{\rho}) = \mathrm{Tr}(f\cdot[h,\rho])$$

or

$$i\hbar\frac{d}{dt}\langle F\rangle = \langle[F,H]\rangle, \tag{12.19}$$

in which the last expectation value was evaluated using Wick's theorem (Sec. C.4). Therefore, we find that all symmetries of the Hamiltonian are conserved, at least on the average. Examples are

(i) the particular number: $(d/dt)\langle N\rangle = (d/dt)\mathrm{Tr}\,\rho = 0$; (12.20)
(ii) the total linear momentum: $(d/dt)\langle\mathbf{P}\rangle = 0$; (12.21)
(iii) the total angular momentum: $(d/dt)\langle\mathbf{J}\rangle = 0$. (12.22)

However, this does not mean that $|\Phi\rangle$ has to be an eigenstate of these symmetry operators. Any localized density distribution $\rho(\mathbf{r})$, for instance, violates translational invariance. We will discuss the problems of symmetry conservation in Section 12.2.2.5.

Furthermore, from (12.19), for velocity independent forces we get for the center of mass $\mathbf{R} = (1/A)\sum\mathbf{r}_i$

$$\frac{d}{dt}\langle\mathbf{R}\rangle = \frac{1}{Am}\cdot\langle\mathbf{P}\rangle. \tag{12.23}$$

12.2.2.4. Conservation of Energy. The derivative of the energy with respect to time can be expressed by the change of the density. Since ρ always stays a Slater determinant, $\dot{\rho}$ only has *ph* matrix elements in the basis in which ρ is diagonal (see Sec. 5.3.3).

$$\frac{d}{dt}\langle H\rangle = \frac{d}{dt}E = \sum_{mi}\left(\frac{\partial E}{\partial\rho_{mi}}\dot{\rho}_{mi} + \frac{\partial E}{\partial\rho_{im}}\dot{\rho}_{im}\right)$$

$$= \mathrm{Tr}(h\dot{\rho}) = \frac{i}{\hbar}\mathrm{Tr}(h\cdot[\rho,h]) = 0. \tag{12.24}$$

This means that the energy is conserved with time. The motion therefore has to take place on lines of constant energy in the multidimensional energy surface of Slater determinants. If, therefore, our initial conditions are inside a valley, such that the energy is smaller than the lowest saddle point, we can never leave that valley. The system runs on equipotential lines within the valley.

In Section 5.4 we discussed an example for such an energy surface in the Lipkin model. The HF energy $\langle H\rangle$ depends here on the two variables α and φ. One can transform them to a coordinate and a canonically conjugate momentum in such a way that the TDHF equations are classical equations of motion in the corresponding phase space [KLD 80].

12.2.2.5. Self-Consistent Symmetries. In Section 12.2.2.3 we have seen that symmetries of the exact Hamiltonian H are connected with conservation laws for the expectation values of the corresponding operators. It is one of the essential features of any mean field theory that the wave functions and densities can break these symmetries. With increasing time the deformation or the position of the nucleus may change drastically.

However, as we have seen in the static case, there may be self-consistent symmetries (Sec. 5.5). They occur in the dynamic case also: If S is a unitary symmetry operation which commutes with the Hamiltonian H and with the initial density $\rho(0)$, then ρ stays invariant under this operation for all times. With $\bar{\rho} = S\rho S^+$, we get

$$i\hbar \frac{\partial}{\partial t}(\bar{\rho} - \rho) = \left[h(\bar{\rho}) - h(\rho), \bar{\rho}\right] + \left[h(\rho), \bar{\rho} - \rho\right], \tag{12.25}$$

that is, if the symmetry is conserved $\bar{\rho} = \rho$ at a time t, it is also conserved at time $t + dt$.

This fact has numerical advantages, because the size of the matrices can be reduced. However, in imposing symmetries we restrict the possible motions drastically and they hinder the development of many *correlations*. Even in cases where symmetry breaking has not much influence on the *macroscopic shape* of the nucleus, the internal single-particle motion can change drastically, because self-consistent symmetries restrict the freedom of each individual single-particle orbital. For instance in the case of axial symmetry, each nucleon has to be aligned parallel or antiparallel to the symmetry axis, and the statistical fluctuations are inhibited. In other words, a fixed wall does not heat up the system, and to get the full amount of one-body dissipation we have to break all possible symmetries.

If a continuous symmetry is broken in a solution $\rho(t)$ we have, as in the static case by applying the symmetry operation alone, an infinite set of other solutions. An example is translations of the coordinate system.

12.2.2.6. Time Reversal Invariance. A density distribution $\rho(t)$ which is a dynamical solution of the TDHF equation (12.9) is not time reversal invariant, in the sense that the time reversal operator [Me 61, Chap. XV]

$$T = K_0 e^{-i\alpha_y}, \tag{12.26}$$

where K_0 is the complex conjugation, changes ρ,

$$\rho_T(t) := T\rho(t)T^+ \neq \rho(t), \tag{12.27}$$

because $\rho_T = \rho$ would require $\dot{\rho} = 0$. On the other hand, we have microscopic reversibility, which means that $\rho_T(-t)$ is also a solution to the TDHF equations. In fact, this property is often used to check the accuracy of computer programs that solve Eq. (12.9). However, this is only a microscopic reversibility and does not mean that the system can not be heated up (within this theory).

12.2.3 Quasi-static Solutions

12.2.3.1. Translations. As we saw in the last section, dynamical solutions of the TDHF equations cannot be entirely time-even. In special cases we can already find dynamical solutions of this equation from a solution of the static HF equations by a suitable transformation that breaks the time reversal symmetry.

A simple example of such a case is the Galilei transformation to a system moving with constant velocity $\mathbf{v}$. It is realized by the operator G (see [Me 61, Chap. XV])

$$G(\mathbf{v}, t) = \exp\left(-\frac{i}{\hbar} \mathbf{v} \cdot (mA \cdot \mathbf{R} - \mathbf{P} \cdot t) \right), \tag{12.28}$$

where $\mathbf{R} = (1/A)\sum \mathbf{r}_i$ is the center of mass and $\mathbf{P} = \sum \mathbf{p}_i$ is the total momentum. To this transformation corresponds a matrix in the single-particle space: $g(\mathbf{v}, t) = \exp[-(i/\hbar)\mathbf{v}(m\mathbf{r} - \mathbf{p}t)]$ which transforms the density

$$\rho(t) \to \bar{\rho} = g\rho(t)\, g^{+}. \tag{12.29}$$

$\bar{\rho}$ obeys the equation of motion

$$0 = i\hbar\dot{\bar{\rho}} = g[h, \rho]\, g^{+} - g[\mathbf{vp}, \rho]\, g^{+} = g[h(\rho) - \mathbf{vp}, \rho]\, g^{+}. \tag{12.30}$$

Using Eq. (12.28) and the symmetry property $g\Gamma(\rho)g^{+} = \Gamma(\bar{\rho})$, which holds for velocity-independent and translational-invariant interactions, we again find the TDHF equations in the moving frame

$$0 = i\hbar\dot{\bar{\rho}} = \left[\frac{1}{2m}(\mathbf{p} + m\mathbf{v})^2 + \Gamma(\bar{\rho}) - \mathbf{v}(\mathbf{p} + m\mathbf{v}), \bar{\rho} \right] = \left[h(\bar{\rho}), \bar{\rho} \right]. \tag{12.31}$$

We thus see that the static solution $[h(\bar{\rho}), \bar{\rho}] = 0$ in the moving frame transforms into a nontrivial dynamic solution of the TDHF equation in the laboratory frame. It takes the form

$$\rho(t) = e^{(i/\hbar)m\mathbf{v}\mathbf{r}} \left(e^{-(i/\hbar)\mathbf{v}t\mathbf{p}} \bar{\rho}\, e^{(i/\hbar)\mathbf{v}t\mathbf{p}} \right) e^{-(i/\hbar)m\mathbf{v}\mathbf{r}}, \tag{12.32}$$

which shows that the stationary solution $\bar{\rho}$ undergoes first a time-even translation of the coordinates by the amount $-\mathbf{v}t$ and then acquires a velocity by a time-odd transformation of the momenta. In the coordinate space we get

$$\rho(\mathbf{r}, \mathbf{r}', t) = e^{(i/\hbar)m\mathbf{v}(\mathbf{r} - \mathbf{r}')} \cdot e^{-\mathbf{v}t(\nabla_{\mathbf{r}} + \nabla_{\mathbf{r}'})} \bar{\rho}(\mathbf{r}, \mathbf{r}'). \tag{12.33}$$

At time $t = 0$ the diagonal density $\rho(\mathbf{r}, \mathbf{r}, 0)$ is identical to the stationary value $\bar{\rho}$. The collective velocity shows up only by $\mathbf{r}$-dependent phase factors in the non-diagonal part.

In practical applications of the TDHF theory to heavy ion reactions, one often uses as initial condition a sum of two densities of the form (12.33) with velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ at a distance such that they do not overlap. To a good approximation they represent two moving ions in their ground state before a reaction takes place.

12.2.3.2. Rotations. In the rotational case we would also like to transform to a rotating system in which the motion can be described by a static solution. However, unlike translations, rotational collective motion is nontrivial. There is no "center of mass angle" and no Galilean invariance in this case (see Sec. 11.3.3). Nevertheless, we can carry out the first part of the transformation (12.28), namely a transformation of the coordinate to a frame rotating with the angular velocity ω

$$\tilde{\rho} = e^{i\omega \mathbf{j}t} \rho(t) e^{-i\omega \mathbf{j}t}. \tag{12.34}$$

In analogy to Eq. (12.30), we get

$$i\hbar \dot{\tilde{\rho}} = e^{i\omega \mathbf{j}t} \left[t + \Gamma(\rho) - \omega \mathbf{j}, \rho \right] e^{-i\omega \mathbf{j}t} = \left[h(\tilde{\rho}) - \omega \mathbf{j}, \tilde{\rho} \right]. \tag{12.35}$$

We now have the stationary HF problem of the self-consistent cranking model (see Sec. 7.7 and 11.4.6.3)

$$\left[h(\tilde{\rho}) - \omega \mathbf{j}, \tilde{\rho} \right] = 0. \tag{12.36}$$

It gives us a density $\tilde{\rho}$ with no time dependence but which contains time-odd components. In this sense it is no static density. Again,

$$\rho(t) = e^{-i\omega \mathbf{j}t} \tilde{\rho} e^{i\omega \mathbf{j}t} \tag{12.37}$$

is a nontrivial solution of the TDHF equations (12.9).

Equation (12.36) can also be derived more fundamentally from a projected TDHF theory in the following manner [EMR 80b]. In analogy to Eq. (12.14), we start with the Lagrangian

$$\mathfrak{L} = \langle \Phi(t) | \hat{P}^I (i\hbar \partial / \partial t - H) \hat{P}^I | \Phi(t) \rangle,$$

where $\hat{P}^I$ is a projector onto good angular momentum (see Sec. 11.4.6). The variation of the action (12.15) with respect to the time-dependent Slater determinant $|\Phi(t)\rangle$ yields projected TDHF equations. In the limit of large deformations we can use the Kamlah expansion (11.80) up to first order and obtain Eq. (12.35). Its simplest solution is given by the static problem (12.36).

If one believes in the Cranking model, one sees in this way that the projection is a very important feature in the time-dependent HF case. This also becomes evident in other problems as, for instance, in the calculation of cross sections, where we have to project on individual reaction channels (see Sec. 12.2.4).

12.2.4 General Discussion of the TDHF Method

At a first look, the TDHF method seems to have appealing *advantages*. It provides a unified description of all kinds of *collective motions* in the nucleus, beginning with the static solution for the ground state and going on over rotational and vibrational excitations to large amplitude collective motion, such as fission, fusion, compound nucleus formations, and deep-

inelastic reactions. The system determines for itself, its path on the multi-dimensional energy surface, which includes all configurations representable by product states. In particular, we are not forced to choose a priori some collective coordinates, nor to force the system to a motion within unphysical external fields. We need not know anything about the flow pattern in the system, or the inertial parameters and their coordinate dependence.

On the other hand, the concept does not only describe collective phenomena: The wave function is a Slater determinant at all times. It takes care of all *single-particle structure* aspects and includes the full interplay between collective and single-particle motion.

Finally, a solution of the TDHF equation also includes friction and *damping* effects insofar as they are produced by the interaction of the particles with the wall of a time-dependent mean field, the so-called *one-body friction* ([BBN 78] and references given there).

To see where this damping comes from in the case of giant resonances we use Thouless' theorem (E.26) and express $\Phi(t)$ in the basis of the HF ground state Φ_0:

$$|\Phi(t)\rangle = \exp\Big(\sum_{mi} z_{mi}(t) a_m^+ a_i\Big)|\Phi_0\rangle$$

$$= |\Phi_0\rangle + \sum_{mi} z_{mi}(t) a_m^+ a_i |\Phi_0\rangle + \sum_{mim'i'} z_{mi}(t) z_{m'i'}(t) a_m^+ a_{m'}^+ a_{i'} a_i |\Phi_0\rangle + \cdots . \tag{12.38}$$

We see that besides the *ph* admixtures (which are already treated in the TDA or RPA in Chap. 8) the TDHF wave functions also contain $2p2h$, $3p3h$ components and so on. To obtain the strength of these admixtures as a function of their energy, we must carry out a Fourier transformation of $|\Phi(t)\rangle$.

Up to now it is unclear how much of the spreading width* of giant resonances comes from the $2p2h$ admixtures of this kind (one-body friction) and how much comes from another kind of $2p2h$ admixtures due to genuine two-body correlations contained in $g^{(2)}$ of Eq. (12.7) (two-body friction). The approximation that the $g^{(2)}$ terms of Eq. (12.7) can be neglected is in turn based on the belief [KK 68] that the mean free path, due to the Pauli exclusion principle, is of the order of the nuclear dimensions. The validity of this assumption certainly breaks down at energies per particle of the order of the Fermi energy, but there is a wide range of phenomena which we should be able to describe by the TDHF method.

In particular, the theory is not necessarily restricted by the adiabatic assumptions—namely, that the wave function is at each time very close to the ground state of the corresponding self-consistent well $h(t)$. For dy-

* We should be aware of the fact, however, that TDHF theory contains the escape width as it is discussed in Section 8.5.4, which means that particles are actually leaving the nucleus.

namic solutions, the TDHF single-particle functions $\varphi_i(t)$, Eq. (12.12), are different from the eigenfunctions in this well:

$$h(t)|\psi_j(t)\rangle = \epsilon_j(t)|\psi_j(t)\rangle. \tag{12.39}$$

This means that we do *not* gain the TDHF Slater determinant $|\Phi(t)\rangle$ by filling the lowest A levels in the potential $h(t)$. The density $\rho(t)$ does not commute with $h(t)$, that is, $\rho(t)$ is not diagonal in the basis of the eigenfunctions $|\psi_i(t)\rangle$ and the occupation probabilities p_j, which are the diagonal elements of ρ in this basis,

$$p_j(t) = \langle\psi_j(t)|\rho(t)|\psi_j(t)\rangle \tag{12.40}$$

are spread over many levels, that is, we obtain a *smeared out Fermi surface* in this basis. On the other hand the distribution of the occupation probabilities should not deviate from a Fermi step function too much, either, that is, it should not become completely fragmentated, otherwise the exclusion principle will no longer suppress two-body collisions. However, no detailed studies on this problem presently exist.

Besides all the great advantages of the TDHF method, however, there are some important *restrictions* which we should be aware of:

(i) The theory provides a fully microscopic description, which follows each single-particle wave function as it changes with time and requires a tremendous amount of *numerical effort*. In particular, we should keep in mind that its full beauty and power can only develop if as many symmetries as possible are broken. Realistic calculations of this type not only go to the limits of modern computer facilities, they also very often make it difficult to extract basic physical features of the system and to understand them in terms of simple models.

(ii) The theory, in a sense, shows a classical *deterministic behavior*: We start with an initial density $\rho(t)$ and follow it and its velocity $\dot{\rho}(t)$ through time in a multidimensional energy surface like a classical trajectory. This fact alone does not say that the theory is a classical one; the exact quantum mechanical statistical operator $\hat{\rho}$ obeys the von Neumann equation, which has a structure very similar to Eq. (12.9). The essential difference from a full quantum mechanical theory is that we restrict our wave function to Slater determinants at all times and end up with a nonlinear equation. We have therefore given up the *superposition principle*, that is, a linear combination of two Slater determinants is (in general) no longer a product wave function. As a consequence of the nonlinearity of our theory we get solutions of a *soliton character* (see Sec. 12.2.5), that is, density distributions which propagate without changing their shape [see, e.g., Eq. (12.32)], whereas the exact quantum-mechanical density $\rho(t) = \langle\Psi(t)|\hat{\rho}|\Psi(t)\rangle$ would certainly show the usual quantum-mechanical dispersion. The fact that the example (12.32) of the static density moving with constant velocity does not show dispersion can also be expressed by the fact that we have destroyed translational invariance in TDHF theory, that is, we are in the intrinsic system of the nucleus and consider its center of mass coordinate and its momentum as two simultaneous measurable

(classical) quantities. This, of course, is linked to considering Slater determinants as an approximation to the exact wave function.

(iii) Another quantum-mechanical effect is *tunneling* through a potential barrier. If we have two separated local minima in the energy surface, and we start with an energy smaller than the lowest saddle between these minima in one valley, we will never reach the other valley because of the conservation of energy [Eq. (12.24)]. It is a consequence of the nonlinearity that there can exist two different valleys. A linear theory always has only one minimum, namely the ground state (see Sec. 5.2). The impossibility of tunneling becomes particularly evident in cases where the set of general Slater determinants is restricted to a product wave function $|\Phi(q)\rangle$ with a few collective degrees of freedom (see Sec. 12.3.3) as in the description of the fission process. In such cases we must either requantize or introduce linear superpositions of product states. This brings us back to the method of generator coordinates (Chap. 10).

Barriers in the space of Slater determinants, which eventually go over into barriers in a collective coordinate, should not be mixed up with barriers in an external field $V(x)$ in coordinate space. For instance, let us study a static solution which moves with a certain velocity towards such a barrier. The wave function $|\Phi\rangle$ of this system consists of a product of single-particle wave functions $\varphi_i(x)$. Each of them has a tail reaching through the barrier. In general, therefore, $|\Phi\rangle$ is split into two parts at this barrier, a transmitted part and a reflected part. Separately, the two parts are no longer Slater determinants, and the single-particle wave functions φ_i belong to both parts. Depending on the incident energy, the transmitted and reflected parts have very different sizes. Figure 12.1a shows the results of a one-dimensional calculation, where a slab of nuclear matter is moving towards a barrier of Gaussian shape with an energy per particle equal to the height of the barrier. We see that the whole slab is transmitted. Without the interaction of the particles through their mean field, we would have expected a transmission probability of 0.5 for each single-particle wave function. The self-consistent field, together with the external barrier, however, gives a drastically reduced effective barrier for the particles. Figure 12.1b shows the same process at a higher barrier. Now the initial slab is split into two parts. We observe again that the collective penetrability is much higher than the penetrability for one particle.

(iv) There exist fundamental questions in the *physical interpretation* of the results contained in the wave function $|\Phi(t)\rangle$. This is already seen in the example of the slab moving towards an external barrier: Does all of the slab penetrate with a certain probability, or does the slab at each collision break into two separate parts with masses corresponding to the integrated density $|\Phi|^2$ on the left and right sides of the barrier? In the latter case we would get non-integer particle numbers in each part, because the scattered fragments, as we said, are no longer Slater determinants by themselves; thus the single-particle wave functions simultaneously belong to all of the fragments. We could also imagine a complicated mixture of these two possibilities. Physically, we are interested in the probability of the transmission of A_1 particles and the reflection of $A_2 = A - A_1$ ones, that is, the probability that there are A_2 particles to the left and A_1 particles to the

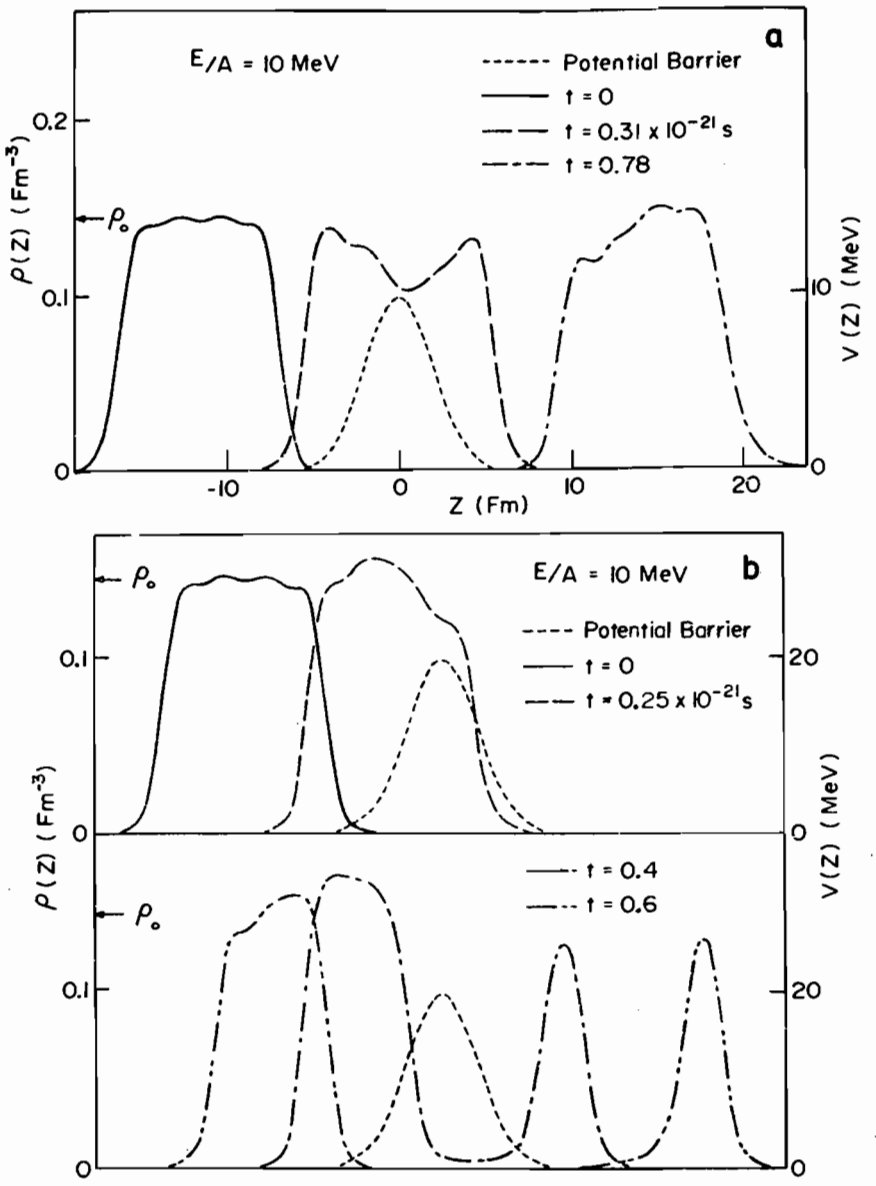


Figure 12.1. Interaction of a slab of nuclear matter with a Gaussian barrier at a height of 10 MeV (a) and 20 MeV (b). (From [BKN 76].)

right of the barrier in the limit of $t \to \infty$:

$$P_{A_1 A_2} = \frac{(A_1 + A_2)!}{A_1! A_2!} \int_0^\infty dx_1 \ldots dx_{A_1} \int_{-\infty}^0 dx_{A_1+1} \ldots dx_A \, |\Phi(x_1 \ldots x_A, t \to \infty)|^2 . \tag{12.41}$$

This is a practical prescription to calculate mass distributions, which has also been applied to calculations for heavy-ion collisions [Ke 76].

In the general case, however, we would like to know the S-matrix elements for the different channels in such a reaction. It is an open question whether we can get such information from a TDHF calculation. In principle, we can consider the wave packet $|\Phi(t)\rangle$ of the TDHF equation to be a valuable approximation to the exact wave packet $|\Psi(t)\rangle$ and so get the S-matrix elements from $|\Phi(t)\rangle$ in quite the same way as we would if we had $|\Psi(t)\rangle$; we could, for example, project $|\Phi\rangle$ onto a model scattering state. However, this would imply calculating the overlap of two different many-body wave functions, which can lead to very erroneous results, as can be seen from the following considerations: Let us assume that we calculate the overlap of two only very slightly different product wave functions, where all the single-particle wave functions in one determinant are different from all those in the other by an amount ϵ:

$$(1-\epsilon)^A \sim e^{-\epsilon A} \tag{12.42}$$

that is, the overlap is exponentially small for large particle numbers. The same would be true if we wanted to calculate, for example, the overlap of a static HF wave function with the exact one, where we can assume that each single-particle orbit is depleted because of correlations due to the probability ϵ [Ne 77]. Though the HF wave function has only very small overlap with the exact wave function, we nevertheless get good results, for example, for binding energies. This is due to the fact that there we need only calculate the expectation value of one- and two-body operators, which involves at most only two single-particle wave functions. The same considerations are true in the time-dependent case: Calculating S-matrix elements from $|\Phi(t)\rangle$ (see above) means that we calculate expectation values of an A-body operator in the case of heavy-ion reactions (the configuration of all A nucleons is changed), and thus it probably gives rather doubtful results. Therefore, in the time-dependent case we must also restrict ourselves to the evaluation of few-body operators such as particle number dispersion or mean excitation energies.

Closely related to the above problems is the question of whether, starting from a Slater determinant as an initial condition, the exact wave function $|\Psi(t)\rangle$ stays close to a Slater determinant in the course of time. To this end, Lichtner and Griffin [LG 76a] investigated the time dependence of the function

$$D(t)=\frac{1}{A}\mathrm{Tr}\left(\rho(t)-\rho^2(t)\right). \tag{12.43}$$

Starting with a Slater determinant $|\Phi_0\rangle$ at $t=0$, we find for short times,

$$D(t)=\left(\frac{t}{\tau}\right)^2+\cdots \tag{12.44}$$

with a limit on the lifetime

$$\frac{1}{\tau}\geqslant\frac{2}{\hbar}\sqrt{\frac{1}{A}\left\{\langle\Phi_0|H^2|\Phi_0\rangle-\langle\Phi_0|H|\Phi_0\rangle^2\right\}}\ , \tag{12.45}$$

which is proportional to the energy spread per particle in the function $|\Phi_0\rangle$. For realistic nuclei, we find τ values between 10^{-22} and 10^{-21} seconds. This means that the wave function $|\Psi(t)\rangle$ already looks quite different to a Slater determinant after very short times. This again shows the fact that the overlap $\langle\Psi(t)|\Phi(t)\rangle$ becomes very small as soon as $|\Psi(t)\rangle$ is only slightly different from $|\Phi(t)\rangle$, but this does not mean that $|\Phi(t)\rangle$ cannot be used for the evaluation of few-body operators.

(v) Perhaps the most serious problem in the TDHF theory is the *spurious final state interaction*. In an exact calculation we would have many different outgoing channels, which do not interact for $t\to\infty$. In our example of a slab moving towards a barrier, we could consider, for instance, the penetration of one, two, or three particles, whereas the rest is reflected. In general, the transmitted systems would have different velocities. Within the TDHF theory, however, all the channels are trapped into one common mean potential on the right-hand side of the barrier. To stay within this potential, all the channels have to exchange momentum permanently. Therefore, for large times we also get an unphysical spurious interaction between the different channels. Even if we do not calculate S-matrix elements, we cannot exclude that this final state interaction does not also give spurious contributions to mean values of few-body operators. In analogy to the symmetry violation in the static HF problem, we should therefore do some kind of *variation after projection* (see Chap. 11) onto the different channels. The TDHF function $|\Phi(t)\rangle$ has to be understood as an "internal" wave function.

In fact, one can derive the TDHF equations from a saddle point approximation in the Feynman path integral representation [FH 65] of the exact quantum mechanical propagator [Kl 77, ER 78, NLP 79, Re 79] where the above questions seem to be treated more properly.

12.2.5 An Exactly Soluble Model

In order to study some of the problems discussed above, Yoon and Negele [YN 77] investigated an exactly soluble model of N bosons interacting through a δ-force in one dimension.* The Hamiltonian is given by

$$H=-\frac{1}{2}\sum_{i=1}^{N}\frac{\partial^2}{\partial x_i^2}-v\sum_{i<j=1}^{N}\delta(x_i-x_j). \tag{12.46}$$

This model has been solved exactly for bound states [Be 31, Mc 65] and for scattering states [Ya 67]. To get the time-dependent Hartree equation for a product state where all the bosons occupy the lowest level with the wave function $\varphi(x,t)$, in a self-consistent field, we use the variation principle (12.16) with

$$\langle\Phi|i\frac{\partial}{\partial t}-H|\Phi\rangle=N\int dx\left(\varphi^* i\frac{\partial}{\partial t}\varphi+\frac{1}{2}\varphi^*\frac{\partial^2}{\partial x^2}\varphi+\frac{v}{2}(N-1)\varphi^*\varphi^*\varphi\varphi\right) \tag{12.47}$$

*A similar, but more complicated, model has been proposed for fermions [NW 76a+78].

and find

$$i\frac{\partial}{\partial t}\varphi+\frac{1}{2}\frac{\partial^2}{\partial x^2}\varphi+v(N-1)\varphi^*\varphi\varphi=0. \tag{12.48}$$

The static solution is

$$\varphi_s(x)=\frac{\sqrt{(N-1)\cdot v}}{2\cosh((N-1)vx/2)} \tag{12.49}$$

with the energy

$$E_H=-\frac{N(N-1)^2v^2}{24} \tag{12.50}$$

and the density

$$\rho_s(x)=\frac{N(N-1)v}{4\cosh^2((N-1)vx/2)}\,. \tag{12.51}$$

In the dynamical case for $2N$ particles Eq. (12.48) has a two-soliton solution [ZS 72, Do 76] (for the definition and properties of solitons, see the review articles [SCM 73, Ra 75]):

$$\varphi(x,t)=\frac{\sqrt{(2N-1)v}}{2}e^{-(i/2)(K^2-a^2)t}$$
$$\times\frac{\left(e^{iKx}\cdot\left\{e^{-a(x-Kt)}+\left[K^2/(K-ia)^2\right]e^{-a(3x+Kt)}\right\}\right)+(K\leftrightarrow -K)}{1+2e^{-2ax}\cosh(2aKt)-2a^2e^{-2ax}\,\mathrm{Re}\left(e^{2iKx}/(K+ia)^2\right)+\left(K^4/(K^2+a^2)^2\right)e^{-4ax}}, \tag{12.52}$$

with $4a=(2N-1)\cdot v$ and K an arbitrary real number. It describes the transmission of two solitary waves through one another with relative momentum $2K$. The wave function is given in the center of mass system. To find the density in the system of one part of the wave, we write $y=x-Kt$ and get

$$\rho(y,t)=2N\cdot|\varphi(y,t)|^2\underset{t\to\infty}{=}\frac{N\cdot(N-\frac{1}{2})v}{4\cosh^2\left((N-\frac{1}{2})vy/2\right)}, \tag{12.53}$$

which, for large N-values, agrees with the static density (12.51). We also can show that the time delay resulting from the interpenetration of the two waves agrees with the exact time delay for large values of N [Do 76]. On the other hand the lifetime of the determinant (12.45) goes to zero for large N. This model therefore gives us an example that results from a mean field theory and may be exact in the limit of large systems, even if the wave function has no overlap with the exact wave function.

12.2.6 Applications of the TDHF Theory

The oldest application of TDHF theory is the description of vibrations around a static solution ρ_0 in the limit of small amplitudes. In Chap. 8 we saw that by linearizing Eq. (12.9) with respect to $\delta\rho=\rho-\rho_0$ we gain the *random phase approximation*.

In the limit of small velocities $\dot{\rho}$ but large amplitudes, we get the *adiabatic-TDHF* theory. This will be discussed with its applications in Section 12.3. We shall therefore restrict ourselves here to exact numerical solutions of the TDHF equations (12.9).

TDHF theory has been applied to *exactly soluble models*. We gave such an example in the last section. Another case is that of the Lipkin model [Kr 77, KLD 80]. In Fig. 5.3 the energy surface of this model is shown. The solutions of the TDHF equations are contour lines of constant energy [KLD 80].

In all the more *realistic calculations* [BKN 76, KDM 77, CMM 78, BGK 78, FKW 78], Skyrme like forces were used. The effective range corrections ($\sim p^2$) were neglected or in some cases replaced by a Yukawa force with finite range and a special exchange part which restricts it to contribute only to the direct term of the self-consistent potential. The exchange term in the Coulomb force is neglected, too. Thus Eq. (12.12) is only a differential equation and not an integrodifferential equation, as it would be in the general case.

In a further simplification we restrict our discussion to *slabs of nuclear matter* which have also been investigated [BKN 76]. This is a one-dimensional problem in the coordinate z:

$$i\hbar\dot{\varphi}_i(z,t)=\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial z^2}+\Gamma(z,t)\right)\varphi_i(z,t)=\hat{h}(z,t)\varphi_i(z,t). \tag{12.54}$$

Discretization in the coordinate z yields a matrix equation

$$i\hbar\dot{\varphi}(t)=h(t)\varphi(t). \tag{12.55}$$

Finally, the time t is also discretized. The obvious way to proceed would be

$$i\hbar\frac{1}{\Delta t}(\varphi^{(n+1)}-\varphi^{(n)})=h^{(n)}\varphi^{(n)} \tag{12.56}$$

or

$$\varphi^{(n+1)}=\left(1-\frac{i}{\hbar}\Delta t\cdot h^{(n)}\right)\varphi^{(n)}. \tag{12.57}$$

Since $(1-(i/\hbar)\Delta t\cdot h^{(n)})$ is not unitary, this yields numerical instabilities. A way to avoid this is to replace Eq. (12.56) by

$$i\hbar\frac{1}{\Delta t}(\varphi^{(n+1)}-\varphi^{(n)})=\frac{1}{2}h^{(n)}(\varphi^{(n+1)}+\varphi^{(n)}) \tag{12.58}$$

or

$$\varphi^{(n+1)}=\frac{1-\frac{i}{2\hbar}\Delta t\cdot h^{(n)}}{1+\frac{i}{2\hbar}\Delta t\,h^{(n)}}\varphi^{(n)}, \tag{12.59}$$

with a unitary approximation to the evolution operator (the so-called Crank–Nicholson operator [CN 47]). This equation, however, involves a matrix inversion, and recently a truncation of the Taylor series of the evolution operator to the first few terms has proved to be satisfactorily accurate [FKW 78].

The force constants were adjusted so as to reproduce the properties of nuclear matter in the static case. In order to investigate the oscillations of an individual

slab, the initial conditions were chosen as

$$\varphi_i(\mathbf{r}, t=0) = e^{iS(\mathbf{r})}\varphi_i^0(\mathbf{r}), \tag{12.60}$$

where φ_i^0 are the static single-particle wave functions. The function $S(\mathbf{r})$ determines the velocity distribution for $t=0$, which is defined by the distribution of the current $j(\mathbf{r}, t)$,

$$\mathbf{v}(\mathbf{r}, t) = \frac{\mathbf{j}(\mathbf{r}, t)}{\rho(\mathbf{r}, t)} = \frac{1}{\rho(\mathbf{r}, t)} \sum_{i=1}^{A} \frac{\hbar}{2mi}(\varphi_i^* \nabla \varphi_i - \varphi_i \nabla \varphi_i^*) \tag{12.61}$$

or

$$\mathbf{v}(\mathbf{r}, t=0) = \frac{\hbar}{m} \nabla S(\mathbf{r}). \tag{12.62}$$

For $S(\mathbf{r}) = \alpha z^2$, we get $v(z) = (2\hbar/m)\alpha z$. With this initial condition the slab starts to oscillate. Figure 12.2 shows such vibrations, which correspond to breathing modes for different values of α. Even for small α these oscillations are not harmonic and they do not seem to be associated with a single frequency. For large α-values the oscillations show drastic fluctuations. Finally, the excitation is too strong and the slab splits into pieces (not shown).

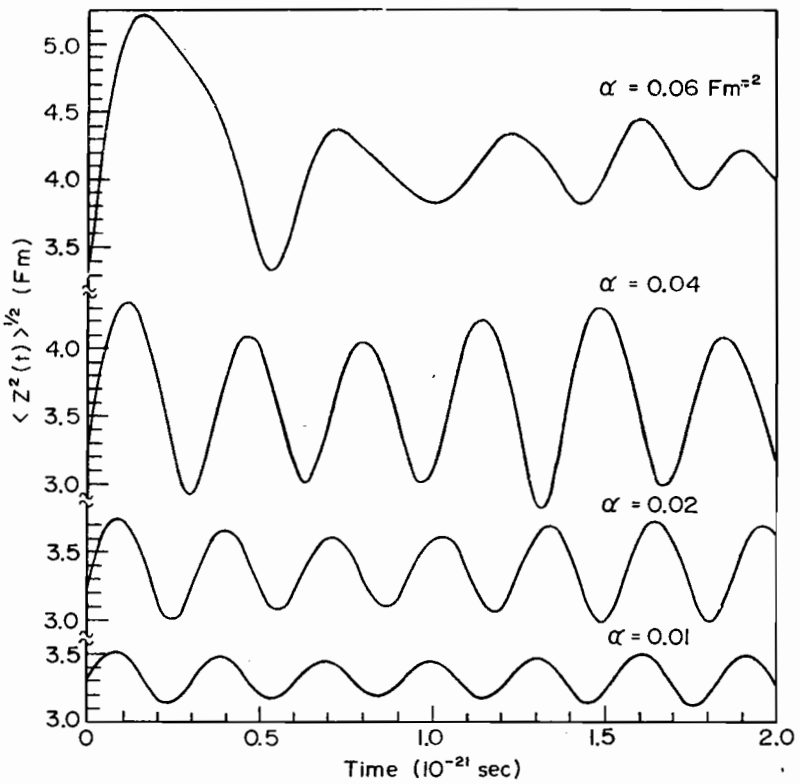


Figure 12.2. Root mean square length of an oscillating slab as a function of time in TDHF approximation for different values of the initial velocity α. (From [BKN 76].)

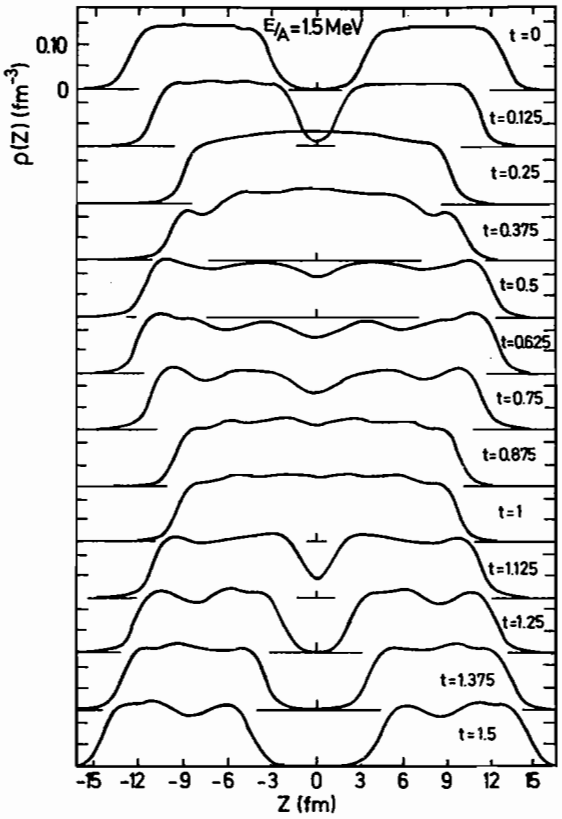


Figure 12.3. Density profiles $\rho(z, t)$ at various times for the scattering of two identical slabs at an intermediate energy. (From [BKN 76].)

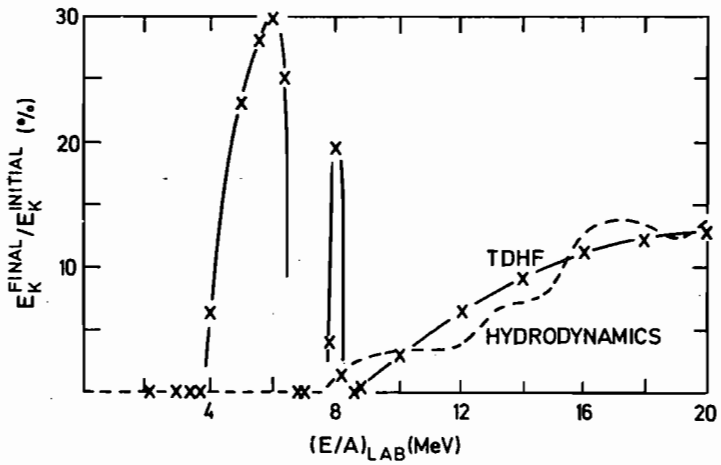


Figure 12.4. The ratio of final translational kinetic energy to initial translational kinetic energy as a function of E/A for a collision of two slabs of nuclear matter. A value of zero denotes fusion. (From [BKN 76].) The dashed line corresponds to a hydrodynamical calculation, discussed in Sec. 13.3.3. (From [WMW 77].)

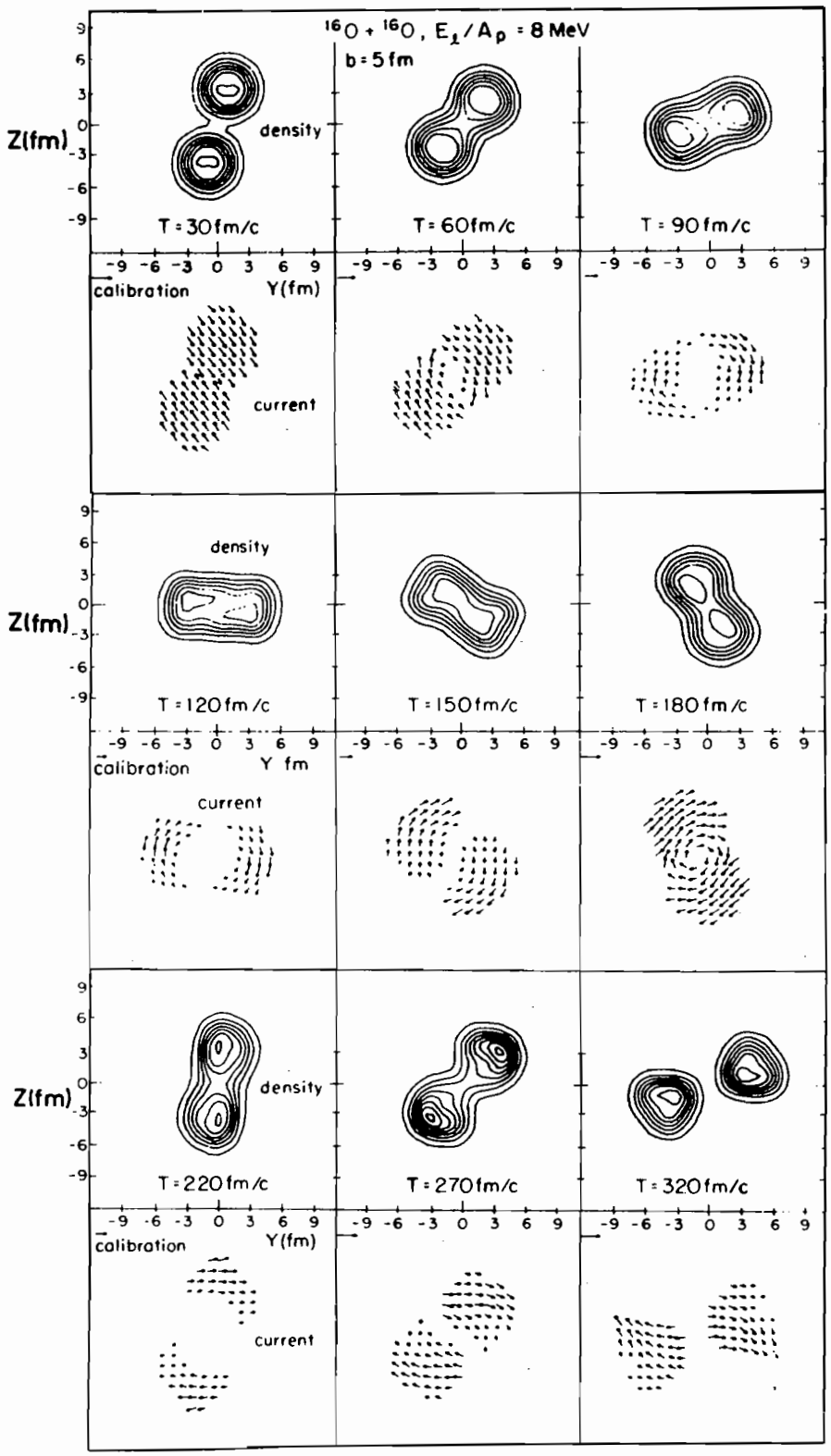


Figure 12.5. Density and velocity distributions in the laboratory system of $^{16}O + {}^{16}O$ at impact parameter $b = 5$ fm at different times T. (From [CMM 78].)

Starting with two self-consistent slabs traveling in opposite directions, we can investigate the collision process. We find a great variety of phenomena. Depending on the bombarding energy, inelastic scattering, compound nucleus formation, resonance behavior, strongly damped collisions, transfer processes, and fragmentations have been observed. Figure 12.3 shows $\rho(z, t)$ at different times for an intermediate energy. After an initial fusion the central density is increased. The compound system begins to oscillate and undergoes fission after one period.

For other energies fusion occurs. As we see in Figure 12.4, we always get fusion for low energies and always a breakup of the compound system for high energies; in between, there is a resonance region. In detail, this depends very much on single-particle motion in the neck region at the scission point. If single-particle effects generate a depletion of this region at the relevant moment of time, scission occurs, otherwise the system oscillates back. One cycle later, much energy has perhaps dissipated in other degrees of freedom and the system stays together. Such resonance structure has actually been observed in the total fusion cross section of heavy-ion reactions [SVE 76].

The slab geometry is not very realistic, since the translational invariance in the x and y directions prevent nuclear matter splashing aside during the bombarding process, and the high degree of symmetry reduces the damping mechanism. Therefore, two-dimensional [Ko 76, CM 76, MC 76], quasi-three-dimensional [KDM 77, MDK 77] and fully three-dimensional [CSM 76, CMM 78, BGK 78, FW 78] calculations have been carried out. They require a great deal of numerical effort and give nice pictures of the density and current distributions in such heavy-ion scattering processes* (see Fig. 12.5).

A TDHF calculation has also been used to describe the fission process [NKM 78]. Starting very close to the saddle point, the system has been followed until scission into two separated fragments has occurred. However, here we do not wish to go into further details of the numerous numerical calculations using TDHF theory.

12.3 Adiabatic Time-Dependent Hartree–Fock Theory (ATDHF)

12.3.1 The ATDHF Equations

As we saw in the last section, nowadays complete numerical solution of the full TDHF equations is possible in some cases. The interpretation of the results, however, is difficult. In this section we will therefore study a theory which makes a further simplifying approximation: Besides the assumption that the wave function stays a Slater determinant for all times (which characterizes the TDHF approach), we investigate only slow processes and

* For an introduction to heavy-ion collisions see [NW 76b].

expand the velocity to second order. However, the aim of this theory is not to solve the TDHF equations (12.9) in some approximation, but to derive in a fully microscopic way a *collective Hamiltonian* for the description of collective processes which involve only small velocities. Thus it provides a connection between a microscopic theory of the nucleus and the phenomenological models discussed in Chapter 1. In particular, in this way we are able to derive microscopic expressions for the parameters of the collective model, such as moments of inertia and energy surfaces. We have already seen in Chapter 10 that the GCM method is able to give such a collective Hamiltonian. The ATDHF theory presents a somewhat different approach.

Since the collective model is quadratic in the velocities, we are restricted to the *adiabatic limit*. This requires that the collective motion is slow compared to the single-particle motion of the nucleus, so that we can neglect all terms of higher than second order in the velocities. As we shall discuss in Section 12.3.5.2, this assumption is justified for a large variety of problems in nuclear physics, such as collective excitations of small amplitude, low-lying anharmonic spectra in transitional nuclei, nuclear fission, and heavy-ion collisions at lower energies. The name "adiabatic" in this context means that the collective *velocity* is small. We often compare the corresponding frequencies, that is, the corresponding Fourier components of the time dependence. For the case of a *large amplitude* motion (for instance, fission) these frequencies have to be small to get a small velocity. For the case of *small amplitudes*, however, the frequencies need not be restricted. We can also have adiabatic motion with rather high frequencies, such as those for giant resonances. In both cases for the kinetic energy, it is sufficient to go only to second order in the velocities and both cases are treatable by the adiabatic approximation. For small amplitudes we can, in addition, stop after second order in the potential energy and obtain the harmonic limit (RPA, see Chap. 8). For large amplitudes this is no longer possible. The ATDHF approach is therefore a theory of collective motions of *arbitrary amplitude* but *small velocity*.

In the last few years there have been numerous attempts to develop approximations along these lines, and the field is still very much under investigation. To be complete within this context goes beyond the scope of this book. Therefore, in the following we shall present the version of *Baranger and Veneroni* [Ba 72a, BGV 76, BV 78] in some detail and show the connection with the work of Villars [Vi 71, 75, 77]. Other important papers in this context are given in the references [Be 65, Be 68, Ba 67, WSR 71, FC 72, He 73b, Ho 73, HY 74, RB 74, Pa 75b, EBG 75, KK 76, Go 76b, RB 76, Ma 77b]. For pedagogic reasons we shall also restrict ourselves to pure Slater determinants ($\rho^2=\rho$), although it is possible and necessary in many practical cases to include pairing correlations.

The theory involves two approximations: (i) the *TDHF assumption*, where the wave functions stay a Slater determinant at all times; and (ii) the *adiabatic approach*, where we must include the velocities only to second order. Within the ATDHF theory the *classical aspect* of the theory is

studied in more detail. It will turn out that we can define classical coordinates and canonically conjugate momenta, which obey classical equations of motion.

The first problem, therefore, consists in defining these *coordinates* and *momenta*. In TDHF theory everything is determined by the single-particle density $\rho(t)$, which is not time-reversal invariant. Since we want to have time-reversal invariant coordinates, Baranger and Veneroni proposed the following decomposition of the density $\rho(t)$.

$$\rho(t) = e^{(i/\hbar)\chi(t)} \rho_0(t) e^{-(i/\hbar)\chi(t)}, \tag{12.63}$$

where ρ_0 and χ are both Hermitian, time-even matrices, that is,

$$\rho_T(t) := T\rho(t)T^+ = e^{-(i/\hbar)\chi(t)} \rho_0(t) e^{(i/\hbar)\chi(t)}. \tag{12.64}$$

Again, ρ_0 satisfies $\rho_0^2 = \rho_0$, $\mathrm{Tr}\,\rho_0 = N$, that is, it corresponds to an N-dimensional Slater determinant $|\Phi_0\rangle$. In the following the ρ_0 shall be the coordinate and as we shall see, χ shall be the momentum.

We are much more familiar with a time-even determinant ρ_0 as coordinate than with the TDHF density ρ, which mixes time-even and time-odd components. In Chapter 2, for instance, we have used Nilsson functions with certain deformation coordinates q_i which correspond to time-even densities. The same is true for CHF calculations (see Sec. 7.6.) with a time-even constraint.

From the property that $\rho_0(t)$ is time-even, it follows that the total current associated with the density $\rho_0(t)$ vanishes for all times:

$$\mathbf{j} = \frac{1}{m} \mathrm{Tr}(\mathbf{p}\rho_0(t)) = 0, \tag{12.65}$$

because the trace of the product of a time-even and a time-odd operator vanishes unless it is purely imaginary.

In the following we therefore call $\rho_0(t)$ a *static* density. It should not be confused with a time-independent density nor with the stationary density, which is a solution of the static HF equation $[h(\rho), \rho] = 0$. However, it will turn out that ρ_0 is the solution of a static HF equation with a suitable (time-dependent) time reversal invariant constraint [see Eq. (12.120b)]. Analogously, we can represent $\rho_0(t)$ by

$$\rho_0(t) = e^{-(i/\hbar)\varphi(t)} \bar{\rho} e^{(i/\hbar)\varphi(t)}, \tag{12.66}$$

where $\varphi(t)$ is a Hermitian, time-odd single-particle operator [Eq. (E.40)] and $\bar{\rho}$ is a time-independent single-particle density (for instance, the solution of the static HF problem $[h(\bar{\rho}), \bar{\rho}] = 0$).

We have already had an example of such decompositions of the density $\rho(t)$ in the case of translations [Eq. (12.32)], where χ is the time-even operator $m\mathbf{v}\mathbf{r}$ and φ is the time-odd operator $t\mathbf{v}\mathbf{p}$. In the case of rotations (12.37), $t\boldsymbol{\omega}\mathbf{j}$ is time-odd, that is, it corresponds to the operator φ. The density $\bar{\rho}$ itself contains time-odd components (see Sec. 7.7) and is time independent. Since there is no center of mass "angle," in this case we are unable to give an explicit expression* for the operator χ.

* It has been calculated, for instance, in RPA order as the solution Q of the Thouless–Valatin equation (8.106).

In the following we will only work with the decomposition (12.63). It can be shown (see Sec. D.2) that it is unique under the following conditions.

(i) χ has only eigenvalues χ_μ with

$$-\tfrac{\pi}{4}\hbar \leqslant \chi_\mu < \tfrac{\pi}{4}\hbar. \tag{12.67a}$$

(ii) The *pp* and *hh* matrix elements of χ vanish in the basis in which ρ_0 is diagonal, that is,

$$\chi^{hh} = \rho_0 \chi \rho_0 = 0; \qquad \chi^{pp} = \sigma_0 \chi \sigma_0 = 0. \tag{12.67b}$$

Here we have used the fact that ρ_0 is the projector onto hole states and $\sigma_0 = 1 - \rho_0$ is the projector onto particle states in the basis in which ρ_0 is diagonal. In the following, we will always work in this basis. It should be noticed, however, that the basis depends on the time since $\rho_0(t)$ is a function of time.

We might ask whether the conditions (12.67) are necessary. In fact, this is not the case—it is only a quite natural choice, * because the *pp* and *hh* matrix elements play a rather important role. Baranger and Veneroni [BV 78] have shown that any other decomposition with non-vanishing matrix elements χ^{pp} and χ^{hh} gives results which deviate from the ones we will derive in the following only to orders in χ that are neglected in the ATDHF approach, anyway. It is therefore consistent with this approximation to use the conditions (12.67), and we shall adopt them in the following.

The *adiabatic approximation* now consists in assuming that the density $\rho(t)$ of the system is at all times close to the density $\rho_0(t)$, that is, we have at all times a nearly static density. In other words, it means that the matrix χ which introduces time-odd components is small. This is certainly not true for all solutions $\rho(t)$ of the TDHF equations, but we are only interested in the ones for which this holds. We can thus expand $\rho(t)$ up to second order in χ and get

$$\rho = e^{(i/\hbar)\chi} \rho_0 e^{-(i/\hbar)\chi} \simeq \rho_0 + \rho_1 + \rho_2 \tag{12.68}$$

with

$$\rho_1 = \frac{i}{\hbar}[\chi, \rho_0], \qquad \chi = -i\hbar[\rho_1, \rho_0], \tag{12.69}$$

$$\rho_2 = \frac{-1}{2\hbar^2}[\chi, [\chi, \rho_0]] = \frac{1}{\hbar^2}\left(\chi\rho_0\chi - \frac{1}{2}(\chi^2\rho_0 + \rho_0\chi^2)\right). \tag{12.70}$$

Equation (D.32) shows that ρ_1 has only *ph*- and *hp*- and that ρ_2 has only *pp*- and *hh*-matrix elements:

$$\rho_1 = \frac{i}{\hbar}(\sigma_0\chi\rho_0 - \rho_0\chi\sigma_0), \tag{12.71}$$

$$\rho_2 = \sigma_0\rho_1^2\sigma_0 - \rho_0\rho_1^2\rho_0. \tag{12.72}$$

* It has been shown, however, that this choice is incompatible with the assumption that χ is local in coordinate space [BQ 78] and sometimes a local χ seems to be the natural choice, as we see in the case of translations (12.32).

In the same way we can expand*

$$h(\rho)=t+\mathrm{Tr}_1(\bar{v}\rho)=h_0+\Gamma_1+\Gamma_2 \tag{12.73}$$

with

$$h_0=t+\mathrm{Tr}_1(\bar{v}\rho_0), \qquad \Gamma_1=\mathrm{Tr}_1(\bar{v}\rho_1), \qquad \Gamma_2=\mathrm{Tr}_1(\bar{v}\rho_2). \tag{12.74}$$

In the expansions of ρ and of h the even terms are always even under time reversal and the odd are always odd.

If we insert the expansions (12.68) and (12.73) into the TDHF Eq. (12.9) we can decompose it according to its behavior under time reversal and get the two *ATDHF equations*:

$$\text{(I)} \qquad i\hbar\dot{\rho}_0=[h_0,\rho_1]+[\Gamma_1,\rho_0] \tag{12.75a}$$

$$\text{(II)} \qquad i\hbar\dot{\rho}_1=[h_0,\rho_0]+[\Gamma_1,\rho_1]+[\Gamma_2,\rho_0]. \tag{12.75b}$$

As we shall see later, only the *ph* and *hp* parts of both equations are important. In deriving these equations we have neglected higher-order terms in χ. To be able to do that, we have to know the order in χ of the time derivatives $\dot{\rho}_0, \dot{\rho}_1, \dot{\rho}_2 \ldots$. In a Fourier analysis they carry a factor ω, where ω is the corresponding frequency. Now we have to distinguish between large amplitude motion and small amplitude oscillations. In the large amplitude case, ω has to be of the order of χ and small; as we have seen, that means $\dot{\rho}_i$ is an order higher in χ than in ρ_i. In the small amplitude case ω does not necessarily have to be of the same order as χ, that is, $\dot{\rho}_i$ is of the same order in χ as ρ_i.

In Eq. (I) above we have therefore neglected all terms that are equal to or of higher order than second and in Eq. (II) all terms that are equal to or higher than third order in χ. In addition, the term $[h_0,\rho_2]$ has been neglected in Eq. (II) for reasons which we shall discuss later.

In zeroth order in χ we gain the static case

$$\dot{\rho}_0=0, \qquad [h_0,\rho_0]=0. \tag{12.76}$$

Equations (I) and (II) therefore always take into account the first nontrivial order in χ. In particular, this shows that in the large amplitude case, where $\dot{\rho}_1$ is of second order in χ, the term $[h_0,\rho_0]$ in (12.75b) is also at least of second order, in spite of the fact that h_0 and ρ_0 separately are large. This means that the static HF equation (12.76) is *almost* satisfied. The nucleus is almost in static equilibrium at all times and the motion is slow.

As we shall see in detail in Section 12.2.2, we can give a *classical interpretation* to the ATDHF equations, in which ρ_0 and χ are canonical conjugate coordinate and momentum. Equation (I) gives a linear connection between velocity and momentum, which depends on ρ_0, that is, the corresponding mass is coordinate dependent. Equation (II) has the time derivative of the momentum on the l.h.s. The r.h.s. therefore represents a force. It depends on the term $[h_0,\rho_0]$ and on further terms quadratic in the momentum χ. Starting from some initial value ρ_0 at zero velocity, $[h_0,\rho_0]$ is therefore the force which accelerates the system at the beginning. Since the momenta stay small for all times, the smallness of $[h_0,\rho_0]$ for all times is

* The abbreviation Tr_1 is defined in Eq. (E.19).

therefore a necessary and sufficient condition for the existence of an adiabatic motion. If there is a whole region in the space of time-even densities ρ_0 in which $[h_0, \rho_0]$ is small, then we can find a trajectory $\rho(t)$ in this region which corresponds to adiabatic motion.

We still have to show that Eqs. (I) and (II) should only be used in the *ph* and *hp* space: The *pp*- and *hh* matrix elements of Eq. (I) are at least of second order in χ. For instance, from Eq. (D.32) $\dot{\rho}_0$ has only *ph* and *hp* matrix elements, and from Eq. (12.71) the same holds for ρ_1. The only non-vanishing *hh* matrix elements of the r.h.s. of Eq. (I) are therefore

$$\rho_0[h_0, \rho_1]\rho_0 = -\rho_0([h_0, \rho_0]\rho_1 + \rho_1[h_0, \rho_0])\rho_0, \tag{12.77}$$

which is at least of second order, since $[h_0, \rho_0]$ is of second order in the large amplitude case and of first order in the small amplitude case. The same holds true for the *pp* matrix elements of Eq. (I).

In Eq. (II) we have left out the term $[h_0, \rho_2]$. From Eq. (12.72) we see that ρ_2 only has *hh* and *pp* matrix elements. The *ph* matrix elements of $[h_0, \rho_2]$ therefore contain only *ph* and *hp* matrix elements of h_0, which are of at least first order in χ. Therefore, $[h_0, \rho_2]$ can be neglected in the *hp* and *ph* part of Eq. (II). In the *hh* and *pp* parts of Eq. (II) we can no longer neglect this term, since the *pp* and *hh* matrix elements of h_0 are large. We can show, however, that the *pp* and *hh* matrix elements of Eq. (II) with inclusion of the term $[h_0, \rho_2]$ vanish identically. This requires a longer calculation [BV 78]; here we give only the most important steps. First we express ρ_2 by Eq. (12.72); then we use Eq. (12.71) and get for the *hh* matrix element of $\dot{\rho}_1$

$$\rho_0\dot{\rho}_1\rho_0 = -\rho_0(\rho_1\dot{\rho}_0 + \dot{\rho}_0\rho_1)\rho_0. \tag{12.78}$$

Finally, we insert Eq. (I) for $\dot{\rho}_0$ and find that the *pp* and *hh* parts of Eq. (II) vanish.

The set of ATDHF equations (12.75) allow us to calculate $\rho_0(t)$ and $\rho_1(t)$ at all times if we start with some initial values $\rho_0(0), \rho_1(0)$ or $\rho_0(0)$ and $\dot{\rho}_0(0)$. This is, however, a problem, which is at least as difficult as the exact TDHF equations, and nobody will do that in practice. The main advantage of the adiabatic theory lies in the fact that it allows us to derive a classical Hamiltonian for the description of the collective motion.

12.3.2 The Collective Hamiltonian

We are now able to express the total energy E of the system by the variables ρ_0 and ρ_1 or ρ_0 and χ. From Eq. (5.28), we find

$$\begin{aligned} E &= \mathrm{Tr}(t\rho) + \tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1(\rho\bar{v}\rho) \\ &= K + V, \end{aligned} \tag{12.79}$$

where we collect into the kinetic energy K all terms which depend on the velocity in second order and in the potential energy V all terms which contain only the static density ρ_0. Linear terms in ρ_1 vanish because of the time-reversal invariance of the Hamiltonian. We get for V the static HF

energy of the density distribution ρ_0:

$$V = \mathrm{Tr}(t\rho_0) + \tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1(\rho_0\bar{v}\rho_0). \tag{12.80}$$

The kinetic energy assumes the form

$$K = \mathrm{Tr}(h_0\rho_2) + \tfrac{1}{2}\mathrm{Tr}_1\mathrm{Tr}_1\rho_1\bar{v}\rho_1. \tag{12.81}$$

It depends quadratically on ρ_1 (or χ), and because of Eq. (12.75a) also quadratically on the velocity $\dot{\rho}_0$. We can write it in several ways. First we use a basis in which the *pp* part as well as the *hh* part of h_0 is diagonal. This is always possible, since ρ_0 is not changed by such a transformation. We call the eigenvalues ϵ_m (particles) and ϵ_i (holes). Using Eqs. (12.69) and (D.35) and the matrices

$$A_{minj} = (\epsilon_m - \epsilon_i)\delta_{nm}\delta_{ij} + \bar{v}_{mjin}, \qquad B_{minj} = \bar{v}_{mnij}, \tag{12.82}$$

we find with $\chi_{im} = \chi^*_{mi}$

$$K = \frac{1}{2\hbar^2}(\chi^*\chi)\begin{pmatrix} A & -B \\ -B^* & A \end{pmatrix}\begin{pmatrix} \chi \\ \chi^* \end{pmatrix}, \tag{12.83}$$

where the index of the matrices runs over all ph pairs (mi) in $(\chi^*_{mi}\chi_{mi})$. It looks very similar to the RPA notation in Chapter 8. It is, however, more general, since the single-particle energies and χ depend on time.

We can also write the ATDHF equation (I) in this notation:

$$\begin{pmatrix} \dot{\rho}_0 \\ \dot{\rho}_0^* \end{pmatrix} = \frac{1}{\hbar^2}\begin{pmatrix} A & -B \\ -B^* & A^* \end{pmatrix}\begin{pmatrix} \chi \\ \chi^* \end{pmatrix}, \tag{12.84}$$

which connects velocities and momenta and shows that the matrix

$$\mathcal{M} := \hbar^2\begin{pmatrix} A & -B \\ -B^* & A^* \end{pmatrix}^{-1} \tag{12.85}$$

represents a *mass tensor* which depends on the coordinate ρ_0. As is easy to see, it corresponds to the Thouless–Valatin value of the moment of inertia (8.142) in the rotational case. The kinetic energy (12.83) assumes the form

$$K = \frac{1}{2}(\dot{\rho}_0^*\dot{\rho}_0)\mathcal{M}\begin{pmatrix} \dot{\rho}_0 \\ \dot{\rho}_0^* \end{pmatrix} = \frac{1}{2}\mathrm{Tr}(\dot{\rho}_0\chi). \tag{12.86}$$

We have thus derived a *classical Hamilton function*, depending on the coordinate ρ_0 and quadratically on the momentum χ

$$E = H(\chi, \rho_0) = \frac{1}{2}(\chi^*\chi)\mathcal{M}^{-1}\begin{pmatrix} \chi \\ \chi^* \end{pmatrix} + V(\rho_0). \tag{12.87}$$

$V(\rho_0) = \langle\Phi_0|H|\Phi_0\rangle$ is the expectation value of the full Hamiltonian in the static wave function $|\Phi_0\rangle$. The kinetic energy has its origin completely in the time-odd parts of the wave function. As we see from Eq. (12.81), it is made up of two contributions. The first contains the static potential h_0 and the second the time-odd potential Γ_1. Neglecting this time-odd part would mean that we forget about the residual interaction $\bar{v}$ in the mass tensor (12.85), and we would obtain the Inglis formula (see Sec. 8.4.7).

To complete the formalism, we have to show that we can get the ATDHF equation (I) and (II) (12.75) as *Hamilton equations* from the function (12.87). The independent variables are ρ_0 and χ. Infinitesimal changes $\delta\rho_0$ and $\delta\chi$, which have, of course, only *ph* and *hp* matrix elements, produce changes in the energy (12.87). The first part is very simple. For fixed ρ_0 we get

$$\frac{\partial H}{\partial \chi^*} = \mathcal{R}^{-1}\begin{pmatrix}\chi\\ \chi^*\end{pmatrix} = \dot{\rho}_0, \tag{12.88}$$

which shows the first ATDHF equation [which corresponds to Eq. (12.84)] to be identical to the Hamilton equation $\dot{q} = \partial H/\partial p$. The derivation of the second ATDHF equation is more complicated. We first calculate the changes of ρ_1 and ρ_2 by a change $\delta\rho_0$. For fixed χ we get from Eqs. (12.69) and (12.70),

$$\delta\rho_1 = \frac{i}{\hbar}[\chi, \delta\rho_0] \qquad \text{and} \qquad \delta\rho_2 = -\frac{1}{2\hbar^2}[\chi,[\chi,\delta\rho_0]]. \tag{12.89}$$

The change in the energy (12.79) is therefore

$$\begin{aligned}\delta E &= \mathrm{Tr}(\Gamma_2\delta\rho_0 + \Gamma_1\delta\rho_1 + h_0\delta\rho_2) + \delta V\\ &= \mathrm{Tr}\left\{\left(\Gamma_2 + \frac{i}{\hbar}[\Gamma_1,\chi] - \frac{1}{2\hbar^2}[[h_0,\chi],\chi]\right)\delta\rho_0\right\} + \delta V.\end{aligned} \tag{12.90}$$

Since χ and $\delta\rho_0$ have only *ph* and *hp* matrix elements, the only matrix elements of h_0 which contribute to these traces are *ph* and *hp* matrix elements. These are, as discussed in the last section, of at least first order in χ. To be consistent we therefore have to neglect the last term in Eq. (12.90) and find as a second Hamilton equation

$$-\dot{\chi} = \frac{\partial H}{\partial \rho_0^*} = \frac{\partial V}{\partial \rho_0^*} + \Gamma_2 + \frac{i}{\hbar}[\Gamma_1,\chi]. \tag{12.91}$$

Of course, we have to use only the *ph* and *hp* elements of these equations. With

$$\sigma_0\dot{\rho}_1\rho_0 = \frac{i}{\hbar}\sigma_0\dot{\chi}\rho_0, \qquad \rho_0\dot{\rho}_1\sigma_0 = -\frac{i}{\hbar}\rho_0\dot{\chi}\sigma_0, \tag{12.92}$$

and the fact that the derivative of the static energy $\partial\mathcal{V}/\partial\rho_0^*$ with respect to the density ρ_0 gives the *ph* matrix elements of h_0 [see Eq. (5.32)], we see that Eq. (12.91) corresponds exactly to the second ATDHF equation (12.75b).

The second ATDHF equation thus describes the *acceleration of the system*. Besides the static force ($\partial V/\partial\rho_0^*$), there are additional terms that have their origin in the coordinate dependence of the mass tensor (12.85).

We have now established the complete analogy to a classical system using the Hamilton formalism. We could also have done this using the Lagrangian formalism [BV 78].

In the *limit of small amplitudes*, we can derive the RPA in a way different than the usual one (see Sec. 8.4.). We assume that $\rho_0(t)$ stays close to the stationary

solution $\bar{\rho}$ of the static HF problem $[h(\bar{\rho}), \bar{\rho}]=0$,

$$\rho_0(t)=\bar{\rho}+\xi(t), \tag{12.93}$$

and expand the potential $V(\rho_0)$ at the point $\bar{\rho}$ up to second order [see Eq. (7.34)]:

$$V(\rho_0)=V(\bar{\rho})+\frac{1}{2}(\xi^*\xi)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} \xi \\ \xi^* \end{pmatrix}. \tag{12.94}$$

With the *stability matrix* $\mathbb{S}$ [Eq. (7.37)] (curvature tensor) we get the energy

$$E(\chi,\xi)=E_{\mathrm{HF}}+\frac{1}{2}(\chi^*\chi)\mathfrak{M}^{-1}\begin{pmatrix} \chi \\ \chi^* \end{pmatrix}+\frac{1}{2}(\xi^*\xi)\mathbb{S}\begin{pmatrix} \xi \\ \xi^* \end{pmatrix}. \tag{12.95}$$

This is a Hamilton function for coupled oscillators. The corresponding equations of motion are solved by harmonic oscillators of the frequency Ω, and we end up with the eigenvalue problem [Go 59]

$$\Omega^2\mathfrak{M}\mathfrak{R}X=\mathbb{S}X. \tag{12.96}$$

The frequencies $(\hbar\Omega)^2$ are the eigenvalues of the matrix

$$\mathfrak{M}^{-1}\mathbb{S}=\mathfrak{R}\mathbb{S}\mathfrak{R}\mathbb{S}, \qquad \text{with} \quad \mathfrak{R}=\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{12.97}$$

They can be found by diagonalizing $\mathfrak{R}\mathbb{S}$ only, which is exactly the RPA matrix. This is only possible because the inverse of the mass tensor $\mathfrak{M}$ has a structure similar to the curvature tensor $\mathbb{S}$.

12.3.3 Reduction to a Few Collective Coordinates

We come back to the problem of large amplitudes. The system of ATDHF equations (12.75) still contains all degrees of freedom and is much too general for a practical application. In particular, it is not easier than the full TDHF equation (12.9); the adiabatic approximation, however, is also valid for only few trajectories. It may therefore be possible to drastically reduce the number of degrees of freedom.

The selection of the proper collective variables is certainly a great problem. Before we discuss this in more detail, we shall first assume that we have achieved such a reduction and have a family of time-even Slater determinants

$$|\mathbf{q}\rangle=|\Phi_0(\mathbf{q})\rangle\triangleq\Phi_0(\mathbf{r}_1,\ldots,\mathbf{r}_A,q_1,\ldots,q_f) \tag{12.98}$$

with corresponding densities $\rho_0(\mathbf{q})$ which have the property that the solution of the ATDHF problem will always stay within this subset of Slater determinants characterized by the real parameters $\mathbf{q}$; that is, there exists a path $\mathbf{q}(t)$ with

$$\rho_0(t)=\rho_0(\mathbf{q}(t)), \tag{12.99}$$

from which we gain for the velocity

$$\dot{\rho}_0(t)=\dot{\mathbf{q}}\frac{\partial\rho_0}{\partial\mathbf{q}} := -\frac{i}{\hbar}\dot{\mathbf{q}}[\mathbf{P},\rho_0]. \tag{12.100}$$

Since $(\partial/\partial\mathbf{q})\rho_0$ has only *ph* and *hp* matrix elements, this equation defines the corresponding elements* $\mathbf{P}$ of the Hermitian single-particle operators $\hat{\mathbf{P}}=(\hat{P}_1,\ldots,\hat{P}_f)$. The single-particle matrices $\mathbf{P}$ have the elements [see Eq. (D.36ff)]

$$\mathbf{P}_{mi}=\mathbf{P}^*_{im}=-\langle m|\frac{\hbar}{i}\frac{\partial}{\partial\mathbf{q}}|i\rangle;\qquad \mathbf{P}_{mm'}=\mathbf{P}_{ii'}=0 \tag{12.101}$$

in the basis in which ρ_0 is diagonal.

Since we now have an explicit expression for the velocity $\dot{\rho}_0$ in terms of $\dot{\mathbf{q}}$ for the kinetic energy from Eqs. (12.86) and (12.100) we get

$$K=\frac{1}{2}\sum_{\mu,\,\mu'=1}^{f} M_{\mu\mu'}(\mathbf{q})\dot{q}_\mu\dot{q}_{\mu'} \tag{12.102}$$

and the real collective mass tensor

$$M_{\mu\mu'}(\mathbf{q})=\frac{1}{\hbar^2}(P^*-P)_\mu\mathcal{M}\begin{pmatrix} P \\ -P^* \end{pmatrix}_{\mu'}. \tag{12.103}$$

We also can define collective momenta p_μ and find, from Eqs. (12.84) and (12.100),

$$p_\mu:=\sum_{\mu'}M_{\mu\mu'}(\mathbf{q})\dot{q}_\mu=\mathrm{Tr}(\rho_1 P_\mu)=\mathrm{Tr}\left(\chi\frac{\partial}{\partial q_\mu}\rho_0\right) \tag{12.104}$$

and gain for the Hamilton function in the collective variables

$$E=H(\mathbf{p},\mathbf{q})=\frac{1}{2}\sum_{\mu\mu'}M^{-1}_{\mu\mu'}\,p_\mu p_{\mu'}+V(\mathbf{q}) \tag{12.105}$$

with

$$V(\mathbf{q})=V(\rho_0(\mathbf{q}))=\langle\mathbf{q}|H|\mathbf{q}\rangle.$$

It remains to be shown that the collective coordinates $\mathbf{q}$ and momenta $\mathbf{p}$ defined in this way obey equations of motion, which correspond to Hamilton's equations derived from the function (12.105). The first equation is trivial:

$$\frac{\partial H}{\partial p_\mu}=\sum_{\mu'}M^{-1}_{\mu\mu'}p_{\mu'}=\frac{d}{dt}q_\mu. \tag{12.106}$$

In the second case we get, with Eqs. (12.87), (12.90), and (12.104), in an

*These matrices correspond to the matrix p in Eq. (12.32). We now use capital letters for them so that they are not mixed up with the collective coordinates q and p.

obvious shorthand notation [BGV 76],

$$\frac{\partial \mathcal{K}}{\partial q_\mu} = \frac{\partial E}{\partial \rho_0^*}\frac{\partial \rho_0}{\partial q_\mu} + \frac{\partial E}{\partial \chi^*}\frac{\partial \chi}{\partial q_\mu} = -\dot{\chi}\frac{\partial \rho_0}{\partial q_\mu} + \dot{\rho}_0\frac{\partial \chi}{\partial q_\mu}$$

$$= -\dot{\chi}\frac{\partial \rho_0}{\partial q_\mu} - \chi\frac{d}{dt}\frac{\partial \rho_0}{\partial q_\mu} + \sum_{\mu'}\dot{q}_{\mu'}\frac{\partial}{\partial q_\mu}\left(\chi\frac{\partial}{\partial q_{\mu'}}\rho_0\right)$$

$$= -\frac{d}{dt}\left(\chi\frac{\partial \rho_0}{\partial q_\mu}\right) + \sum_{\mu'}\dot{q}_{\mu'}\frac{\partial}{\partial q_\mu}p_{\mu'}$$

$$= -\frac{d}{dt}p_\mu, \tag{12.107}$$

because p_μ and q_μ are independent variables $[(\partial/\partial q_\mu)p_{\mu'}=0]$.

One also often introduces a set of Hermitian operators $\hat{\mathbf{Q}}$ which have only *ph* and *hp* elements defined by

$$\begin{pmatrix} Q \\ Q^* \end{pmatrix}_\mu = \frac{\hbar}{i}\sum_{\mu'} M^{-1}_{\mu\mu'}\begin{pmatrix} A & -B \\ -B^* & A^* \end{pmatrix}^{-1}\begin{pmatrix} P \\ -P^* \end{pmatrix}_{\mu'}. \tag{12.108}$$

They are closely connected to the operator

$$\chi = \sum_{\mu\mu'}\dot{q}_\mu M_{\mu\mu'}Q_{\mu'} \tag{12.109}$$

and allow a simple representation of the mass tensor M:

$$M^{-1}_{\mu\mu'} = (Q^*Q)_\mu \mathfrak{M}^{-1}\begin{pmatrix} Q \\ Q^* \end{pmatrix}_{\mu'} = \frac{1}{\hbar^2}\langle \mathbf{q}|\left[\hat{Q}_\mu, \left[H, \hat{Q}_{\mu'}\right]\right]|\mathbf{q}\rangle. \tag{12.110}$$

They obey the relation

$$\langle \mathbf{q}|\left[\hat{P}_\mu, \hat{Q}_{\mu'}\right]|\mathbf{q}\rangle = (P^* - P)_\mu\begin{pmatrix} Q \\ Q^* \end{pmatrix}_{\mu'} = \frac{\hbar}{i}\delta_{\mu\mu'}, \tag{12.111}$$

which says that $\hat{P}_\mu$ and $\hat{Q}_\mu$ are "*weakly*" *canonical variables*.

The operators $\hat{\mathbf{P}}$ and $\hat{\mathbf{Q}}$ allow us to give a representation of the wave function $|\Phi\rangle$ in the vicinity of an arbitrary point $\mathbf{q}_0$. This "*local*" *representation* shows a close analogy to the case of pure translation (see Sec. 12.2.3.) and will be used in the following sections. The operators $\hat{\mathbf{P}}$ and $\hat{\mathbf{Q}}$ act like infinitesimal generators for the wave function $|\Phi\rangle$ in the vicinity of a point $\mathbf{q}_0$. If we know the "momenta" $\mathbf{p}$ (12.104) and the "coordinates" q we get for the ATDHF function $|\Phi\rangle$ at a point $\mathbf{q}_0+\mathbf{q}$

$$|\Phi(\mathbf{q}_0+\mathbf{q})\rangle = e^{(i/\hbar)\mathbf{p}\hat{\mathbf{Q}}}e^{-(i/\hbar)\mathbf{q}\hat{\mathbf{P}}}|\Phi_0(q_0)\rangle \tag{12.112}$$

and

$$\rho(t) = e^{(i/\hbar)\mathbf{p}(t)\mathbf{Q}} e^{-(i/\hbar)\mathbf{q}(t)\mathbf{P}} \rho_0(q_0) e^{(i/\hbar)\mathbf{q}(t)\mathbf{P}} e^{-(i/\hbar)\mathbf{p}(t)\mathbf{Q}}, \quad (12.113)$$

that is, we can obtain the wave function in the vicinity of $\mathbf{q}_0$ (i.e., locally) by a shift in coordinate generated by the operators $\hat{\mathbf{P}}$ and a shift in the momentum generated by the operators $\hat{\mathbf{Q}}$. These operators depend in general on the position $\mathbf{q}_0$. An example, where the $\hat{P}$'s and $\hat{Q}$'s are constant operators, is given by a Galilei transformation [see Eq. (12.32)].

In this case, we see that $\hat{\mathbf{Q}}$ is the center of mass coordinate and $\hat{\mathbf{P}}$ the total linear momentum. The tensor of inertia is a multiple of the unity and its value is just the total mass $A \cdot m$ of the nucleus. Unlike in GCM with one real generator coordinate q (see Sec. 10.7), we now get in the ATDHF approach the proper inertial parameters. The reason for that is that the ATDHF method is a dynamical theory which allows for time-odd components in the wave function (see also Sec. 11.4.5).

Starting from an arbitrary set of time-even determinants $|\mathbf{q}\rangle$, which depend on a set of parameters $\mathbf{q}$, we developed a theory which uses these parameters as collective coordinates. We also defined corresponding momenta such that the p's and q's fulfill the classical equations of motion derived from a Hamilton function in these variables.

It is certainly a major disadvantage of this theory that we end up with a *classical Hamiltonian*. As in the case of the collective model in Chapter 1, we have to requantize this theory to get a collective Hamilton operator, which allows us to calculate wave functions. The requantization always involves some arbitrariness. There are several methods. One possibility is the Pauli quantization [Eq. (1.53)]. Goeke and Reinhard [GR 78] proposed a different method, which has the advantage that it reproduces the same zero-point corrections $\epsilon_0(\mathbf{q})$ of the energy surface as the GCM theory [see Eq. (10.136)]. However, before we discuss why such a quantization is necessary, we want to present a method that determines the optimal family $\rho_0(\mathbf{q})$ from which we should start.

12.3.4 The Choice of the Collective Coordinates

In deriving the Hamiltonian (12.105) we have assumed that we could restrict ourselves to a few collective coordinates $q_1, \ldots, q_f$. The question arises of how to determine these coordinates (i.e., the corresponding wave functions) and how many of them are needed. Of course, we would like to have as few as possible. Since we have seen that the solution of the ATDHF Eqs. (12.75) with certain initial conditions corresponds to a one-dimensional path in the multidimensional energy surface, it seems that one coordinate q would be sufficient if it is properly chosen. The whole concept of deriving a collective Hamiltonian in this coordinate, however, is only meaningful, if the path determined in such a way does not

depend on the initial velocity $\dot{\mathbf{q}}$, that is, we should find more or less the same path $|\Phi_0(\mathbf{q})\rangle$, if we start from different initial velocities. In general, this will not be the case for a one-dimensional path. We could then try using a two-dimensional surface and so on. The number of the necessary collective coordinates $q_1,\ldots,q_f$ therefore depends on the dynamical behavior of the system. The restriction to a few of them is only possible, if there exists a subset $|\Phi(q_1 \ldots q_f)\rangle$ of Slater determinants such that the solution of the ATDHF problem (12.75) stays within this subset for many different initial conditions. This requirement is very similar to the condition (10.24) in the GCM method, in which we had to ask for generating wave functions with the property that the corresponding collective subspace contains eigensolutions of the full Hamiltonian (see also [RG 78]).

In many physical problems it is not clear whether such a set of coordinates exists that describes a collective motion decoupled from all the other degrees of freedom. If this is not the case, we can try to include the other degrees of freedom in an averaged way. This leads to the concept of friction (see, for instance, [BBN 78]). Within this book we do not wish to go into these problems. We shall assume in the following that there is a fixed number of collective coordinates.

Under this condition we are faced with the question of how to determine the optimal set of wave functions $|\Phi_0(q_1 \ldots q_f)\rangle$. There are several ways. The first, which has been used in most of the realistic applications, is *physical intuition*. For instance, we know that for the fission processes we need at least three coordinates, describing the elongation, the neck, and the asymmetry (see Sec. 1.6), and so we choose Slater determinants in a deformed single-particle potential with these parameters. The second way is more sophisticated. We use *constrained Hartree–Fock (CHF) theory* (see Sec. 7.6) with several constraining operators. The physical intuition is then restricted to the optimal choice of these operators. The third way is given by the *ATDHF method* itself and will be discussed in the following. In this way we will find a close connection to the formulation of ATDHF theory as given by Villars [Vi 75, 77].

In deriving the collective Hamiltonian (12.105) we assumed that we knew the set of static wave functions and used the first of the ATDHF equations (12.75a) to calculate the corresponding momenta and inertial parameters. The second equation (12.75b) was only used to derive the equations of motion in the few variables $\mathbf{p}$ and $\mathbf{q}$. This is only a very small part of the information contained in this second equation. Additional information will be used to determine the optimal set of wave functions [GR 78].

In Sec. 7.6 we saw that any Slater determinant $|\Phi_0\rangle$ (with density ρ_0) can be found as a solution of a CHF equation with a suitable constraining operator F, which has only *ph* and *hp* matrix elements in the basis, in which ρ_0 is diagonal:

$$[h_0 - F, \rho_0] = 0. \tag{12.114}$$

In fact, the second ATDHF equation is already of the form of a CHF equation. To make this evident, we shall rewrite it somewhat: As discussed in Section 12.3.1, only the *ph* and *hp* part of this equation is interesting. With (12.69) it can be written as

$$\left[h_0+\dot{\chi}+\Gamma_2+\frac{i}{\hbar}[\Gamma_1,\chi],\rho_0\right]=0, \tag{12.115}$$

which shows that the *ph* matrix elements of F are given by

$$F^{\mathrm{ph}}=\sigma_0\left(-\dot{\chi}-\Gamma_2-\frac{i}{\hbar}[\Gamma_1,\chi]\right)\rho_0. \tag{12.116}$$

We assume that we know χ from the solution of the first ATDHF equation and use Eq. (12.109). The time derivative of χ is then given by*

$$\dot{\chi}=MQ\ddot{q}+\dot{q}^2\frac{d}{dq}(MQ). \tag{12.117}$$

With the equations of motion (12.106) and (12.107) we find

$$\ddot{q}=\frac{d}{dt}\frac{P}{M}=-\frac{1}{M}\frac{dV}{dq}-\frac{\dot{q}^2}{2}\frac{1}{M}\frac{dM}{dq}. \tag{12.118}$$

Since Γ_2 and $[\Gamma_1,\chi]$ are quadratic in $\dot{\mathbf{q}}$, using Eq. (12.116) we gain for the constraining operator F

$$F=\frac{dV}{dq}\cdot Q+O(\dot{q}^2). \tag{12.119}$$

To get a consistent description, the terms $\propto\dot{q}^2$ have to be small and will be neglected. We therefore end up with the possibility of determining the path $\rho_0(q)$ from a constrained HF equation, where the constraint Q is given by Eq. (12.108). The mass parameter $M(q)$ in this equation is determined, for instance, by the condition (12.111). The following set of equations (Villars [Vi 75, Vi 77]) is therefore up to order $\dot{q}^2$ equivalent to the ATDHF equations (12.75)

$$\begin{pmatrix} A & -B \\ -B^* & A^* \end{pmatrix}\begin{pmatrix} Q \\ Q^* \end{pmatrix}=\frac{\hbar}{iM}\begin{pmatrix} P \\ -P^* \end{pmatrix}, \tag{12.120a}$$

$$\left[h_0-\frac{dV}{dq}Q,\rho_0\right]=0, \tag{12.120b}$$

$$\langle\Phi_0|[\hat{P},\hat{Q}]|\Phi_0\rangle=\frac{\hbar}{i}. \tag{12.120c}$$

This system is very similar to that which we get by a self-consistent determination of the path in the generator coordinate method (Sec. 10.8). For a practical application we have to start with an initial guess for the constraining operator in Eq. (12.120b) and calculate the corresponding $\rho_0(q)$. From Eq. (12.100), we can at each point q determine the correspond-

* We restrict ourselves in the following to one collective coordinate q.

ing operator P and from (12.120a) a new constraining operator Q. This procedure has to be continued until convergence is achieved.

Somewhat related is the *local harmonic approach* [HY 74, RB 76, Ma 77b, ABC 77b, MV 77], where P is determined from Eq. (12.120b) in a linear response approximation [see Eq. (8.137)] neglecting curvature corrections [RG 77].

$$\begin{pmatrix} A & -B \\ -B^* & A^* \end{pmatrix}\begin{pmatrix} P \\ P^* \end{pmatrix} = i\hbar C\begin{pmatrix} Q \\ -Q^* \end{pmatrix}, \tag{12.121}$$

where $C=(d^2V/dq^2)$, that is the curvature of our potential. We can then define a collective frequency $\Omega(q)$ [RB 76] by*

$$\Omega(q)=\sqrt{\frac{C(q)}{M(q)}} \tag{12.122}$$

and a boson operator

$$A^+(q)=\sqrt{\frac{M\Omega}{2\hbar}}\left(\hat{Q}-i\frac{1}{M\Omega}\hat{P}\right). \tag{12.123}$$

Since the Eqs. (12.120a) and (12.121) correspond exactly to the RPA equations in the form (8.99), the operator A^+ can be found as the solution of a RPA calculation at each point q (local RPA).

We do not want to go into further details of these and similar methods, which are still under investigation. Summarizing the results of this section, we can say that there are methods to determine an optimal path or an optimal hypersurface (in cases where we need more than one collective coordinate) in the large set of all Slater determinants. We thus obtain the potential and the inertia parameters as a function of the coordinates $\mathbf{q}$, that is, the classical Hamiltonian function (12.105).

However, for a full description of the system, we still need its dynamical behavior. In the classical picture it is given by the functions $\mathbf{q}(t)$. They are obtained from the solution of the classical equations of motion (12.106) and (12.107). In the quantum mechanical case we first have to requantize the Hamiltonian before solving the corresponding Schrödinger equation in collective coordinates.

12.3.5 General Discussion of the ATDHF Method

12.3.5.1 Why Is a Requantization Necessary? The ATDHF method ends up with classical equations of motion of a collective Hamilton function in a few collective variables. In particular, there is no possibility of tunnelling through potential barriers, therefore we cannot for instance describe spon-

* In cases where M or C is negative, the RPA has purely imaginary solutions. Nevertheless, P and Q are well determined and bosons can be defined (for details, see [RB 76] and Sec. 8.4.5).

taneous fission. This only becomes possible if we requantize the collective Hamiltonian. This prescription is not as ad hoc as it might seem since it can be shown [MH 72, BM 78] that the full TDHF theory can be obtained from the exact boson expansion (see Chap. 9) by replacing the boson operators or the corresponding operators for momentum and coordinate by c-numbers. In *this sense*, TDHF as well as ATDHF is a classical theory and it therefore seems natural to requantize. This does not mean, however, that the theory is completely classical; after all, a Slater determinant is a quantum mechanical wave function.

We have already seen in Section 12.2.4 that the restriction to Slater determinants includes some classical features, such as soliton solutions without dispersion. These features are still more pronounced if we restrict ourselves to a few collective variables $\mathbf{q}$. The overlap between two wave functions $|\Phi(\mathbf{q})\rangle$ and $|\Phi(\mathbf{q}')\rangle$ is sharply peaked at $\mathbf{q}=\mathbf{q}'$ if $\mathbf{q}$ is a collective variable (see Sec. 10.7.4), that is, we can write to a good approximation

$$\langle\Phi(\mathbf{q})|\Phi(\mathbf{q}')\rangle \simeq \delta(\mathbf{q}-\mathbf{q}'), \tag{12.124}$$

which says that the functions $|\Phi(\mathbf{q})\rangle$ are wave packets localized in q-space. At the same time, we calculate $\dot{\mathbf{q}}$, which violates the uncertainty relation and is therefore a classical approximation. In a quantum mechanical theory we should use a superposition of many different shapes $|\Phi(\mathbf{q})\rangle$, as we do, for instance, in the GCM method (see Chap. 10).

12.3.5.2 The Validity of the Adiabatic Approximation. We are now able to give a rough criterion for cases of the adiabatic approximation to be valid [BV 78]. We therefore restrict our considerations to one collective variable q (which is, in fact, no restriction, since q could be the exact solution of the ATDHF equations).

The adiabatic approach is equivalent to the fact that the time-odd component in any single-particle wave function

$$e^{i\hat{\chi}}|i\rangle = |i\rangle + i\hat{\chi}|i\rangle \tag{12.125}$$

should have a small norm:

$$|\hat{\chi}|i\rangle|^2 = \sum_m |\langle m|\hat{\chi}|i\rangle|^2 \ll 1 \qquad \text{for all } i.$$

If, as a rough estimate, in Eq. (12.83) we neglect the residual interaction—that is, if we use the Inglis formula for the collective mass—we find for the kinetic energy involved in this motion

$$\mathcal{K} = \sum_{mi} \chi_{mi}^*(\epsilon_m - \epsilon_i)\chi_{mi} \simeq \Delta\epsilon \sum_{mi} |\chi_{mi}|^2, \tag{12.126}$$

where $\Delta\epsilon$ is a typical *ph* energy. If we furthermore assume that the admixtures of odd components is equally distributed over N occupied states, from (12.126) we get the condition for adiabaticity:

$$\mathcal{K} \simeq \Delta\epsilon \cdot N \sum_m |\chi_{mi}|^2 \ll \Delta\epsilon \cdot N. \tag{12.127}$$

This means that the adiabatic approximation is good as long as the collective kinetic energy is small compared to a typical single-particle excitation energy times the number of single-particle states involved in the collective motion. We should note that it is not the quotient of collective energy to single-particle energy which should be small, but that an additional factor N comes into play which measures the collectivity.

This is the reason why we can also use the adiabatic assumption for high-lying collective states in the RPA. As long as there are many particles involved, the collective energy can be much higher than the single-particle excitations. On the other hand, we see that the adiabatic assumption does not work in situations in which only one particle is involved ($N=1$). At isolated level crossings, for instance, the kinetic energy can become of the same order of magnitude as $\Delta\epsilon$ and the adiabatic approximation breaks down. This feature is similar to that already encountered in the momentum expansion of the GCM theory. In cases where many level crossings occur, one should therefore use these methods with extreme care [St 77].

12.3.6 Applications of the ATDHF Method

12.3.6.1. Quadrupole Vibrations for a $Q\cdot Q$-Force. An early version of the ATDHF theory was given in the pairing-plus-quadrupole model (see Sec. 7.4) of Baranger and Kumar for the description of transitional nuclei [BK 68, Ku 74a]. In this case, the theory becomes extremely simple because of the separability of the force. Since the model includes pairing correlations, Baranger and Kumar solved the adiabatic time-dependent Hartree–Fock–Bogoliubov problem. In this chapter we will restrict ourselves to the pure HF case and in the following, therefore, present a version which neglects pairing. We also consider only pure axially symmetric deformations ($Y=r^2Y_{20}$) and neglect exchange terms as discussed in Section 7.4.

The matrix element $\bar{v}$ in this case is given by

$$\bar{v}(1,2)=-kY(1)\cdot Y(2), \tag{12.128}$$

where k is a force constant.

We start according to Section 12.3.3 with a family of static densities $\rho_0(q)$ that are obtained, for instance, from the solution of the CHF problem

$$[h_0(\rho_0)-qY,\rho_0]=0. \tag{12.129}$$

$|m\rangle$, $|i\rangle$ will be a set of simultaneous eigenvectors of ρ_0 and of the pp and hh parts of h_0. We furthermore find that Γ_1 of Eq. (12.74) vanishes.

$$\Gamma_1=-k\cdot Y\cdot \mathrm{Tr}(Y\rho_1)=0 \tag{12.130}$$

because Y is time-even and ρ_1 is time-odd and the trace is real. This produces a tremendous simplification, because in this case the matrix B of Eq. (12.82) vanishes and A becomes diagonal.

From Eqs. (12.84), (12.100), and (12.101) we can calculate the time-odd part:

$$\chi_{mi}=\hbar^2\frac{\langle m|\dot{\rho}_0|i\rangle}{\epsilon_m-\epsilon_i}=\dot{q}\hbar^2\frac{\langle m|\partial/\partial q|i\rangle}{\epsilon_m-\epsilon_i}. \tag{12.131}$$

The matrix elements of $\partial/\partial q$ are obtained from the perturbation theory of Eq. (12.129), which in this case is equivalent to a linear response approach:

$$\langle m|\partial/\partial q|i\rangle = \frac{\langle m|Y|i\rangle}{\epsilon_m - \epsilon_i}. \tag{12.132}$$

This, together with Eq. (12.103), gives the Inglis formula for the mass:

$$M(q) = 2\hbar^2 \sum_{mi} \frac{|\langle m|\partial/\partial q|i\rangle|^2}{\epsilon_m - \epsilon_i} = 2\hbar^2 \sum_{mi} \frac{|\langle m|Y|i\rangle|^2}{(\epsilon_m - \epsilon_i)^3}. \tag{12.133}$$

From the collective Hamiltonian, therefore, we obtain:

$$\mathcal{H} = \tfrac{1}{2} M\dot{q}^2 + V(q), \tag{12.134}$$

where

$$V(q) = \mathrm{Tr}(\epsilon_0 \rho_0(q)) - \tfrac{1}{2} k\, \mathrm{Tr}(Y\rho_0(q)) \cdot \mathrm{Tr}(Y\rho_0(q)). \tag{12.135}$$

Baranger and Kumar [BK 68] obtained the same collective Hamiltonian with minor differences. They took the pairing-plus-quadrupole model Hamiltonian and in this way derived a collective Hamiltonian depending on the five quadrupole deformation parameters $\alpha_{2\mu}$ and the pairing gap Δ. Since they are not interested in pairing vibrations, they restricted Δ to be the solution of the static BCS equations for each value of the five other coordinates.

After a transformation to intrinsic coordinates (as discussed in Sec. 1.5) they ended up with a classical Bohr Hamiltonian (1.47)

$$\mathcal{H} = \frac{1}{2}\left(\sum_i \mathcal{I}_i \omega_i^2 + B_{\beta\beta}\dot{\beta}^2 + B_{\beta\gamma}\dot{\beta}\dot{\gamma} + B_{\gamma\gamma}\dot{\gamma}^2 \right) + V, \tag{12.136}$$

where the seven functions $\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3, B_{\beta\beta}, B_{\beta\gamma}, B_{\gamma\gamma}$, and V depend on the variables β and γ. They are calculated microscopically. The inertial parameters correspond to the Belyaev formula (3.93). This Hamiltonian has been requantized as discussed in Sec. 1.5. Energy levels and wave functions were then calculated numerically [BK 67b, 68].

In this method the coupling between rotations and vibrations and the mixing between different phonon states is fully taken into account. In this way Baranger and Kumar investigated the transition region around the osmium and the samarium isotopes [KB 68, Ku 74a] and found good agreement with experimental data. In particular, they obtained strong deviations from the rotational picture at the low-A end of the W-Os-Pt-transition region and large deviations from the phonon model at the upper end. Their collective wave functions are often smeared out over all possible shapes.

So far, only quadrupole shapes have been used. From Eqs. (12.120) and (12.132) we see that the constraint used in the next step of the procedure, as discussed in Section 12.3.4, would be

$$Q_{mi} = \frac{\hbar^2}{M} \frac{\langle m|Y|i\rangle}{(\epsilon_m - \epsilon_i)^2}. \tag{12.137}$$

Only in cases where the *ph* energies are almost degenerate, this corresponds to the original constraining operator Y.

12.3.6.2. Further Applications. The ATDHF theory has also been applied to *exactly soluble models* such as the Lipkin model of Sec. 6.2 [KG 74] and a three-level

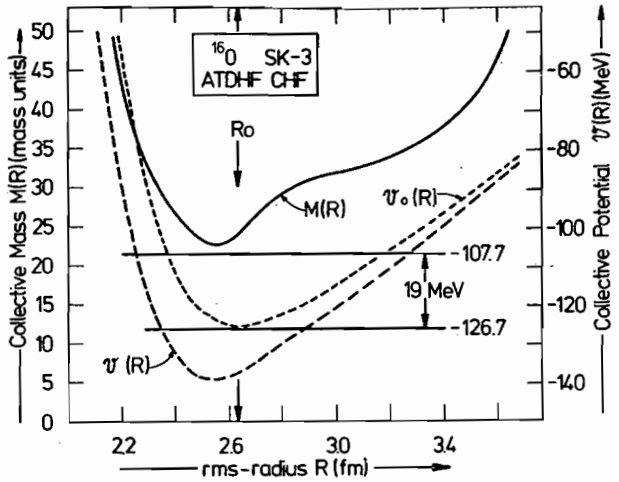


Figure 12.6. Potential energy surface and mass parameter of the isoscalar monopole mode in ^{16}O within ATDHF theory. (From [Go 77].)

Lipkin model [MV 78], which was introduced in [YKD 70]. In general, for cases of large particle numbers—that is, collective situations—the agreement with the exact result is very good.

More *realistic calculations* have been carried out with the Skyrme force [EBG 75, Va 75, Go 77]. In this case, it is convenient to formulate the ATDHF theory in coordinate space. Since time reversal symmetry is broken in the time-dependent theories, in addition to the densities ρ, τ and **J** of Eqs. (5.84f.) we need further densities and currents. In this form the theory has been applied to isoscalar monopole vibrations in ^{16}O and ^{40}Ca using the mass rms radius R as collective coordinate. Figure 12.6 shows the collective mass $M(R)$, the collective potential $v_0(R) = V(R)$, the CHF energy, and the potential $v(R)$, which is obtained by subtracting the zero-point corrections ϵ_0 of Eq. (10.136). The positions of collective eigenstates are indicated.

Giannoni et al. [GMQ 76, GQ 80] have calculated mass parameters for the isoscalar quadrupole modes in light nuclei. They find that the ATDHF mass (12.110) is very close to the simple cranking mass [see Eq. (12.146)] for Skyrme forces with an effective mass $m^*/m \approx 1$. In fact, it can be shown that the term Γ_1 in Eq. (12.75a), which produces the difference between the ATDHF mass (Thouless–Valatin mass), and the cranking mass (Inglis formula) is proportional to $1 - m^*/m$.

12.3.7 Adiabatic Perturbation Theory and the Cranking Formula

The oldest method of deriving the inertia parameters for a collective motion microscopically is the so-called *cranking formula*. It was introduced by Inglis [In 54, 56] in the case of rotations, and we have discussed it in great detail in Section 5.4. Kerman [Ke 61] has derived a similar expression for the mass parameters of more general collective motions. We have seen

in Section 12.3.2 that we obtain this formula again if we neglect the time-odd part Γ_1 of the self-consistent potential (12.74) in the kinetic energy (12.81).

This formula has been used very often in the past. We therefore want to give here the usual derivation in the context of adiabatic perturbation theory (see, for instance, [Sch 68, Di 71]). In particular, we will discuss the validity of the basic assumption of this approximation.

We consider a system of particles in a time dependent, but time even, Hamiltonian $H(t)$. The time dependence is determined by one (or several) collective shape parameter(s) $q(t)$. An example would be a single-particle Nilsson Hamiltonian (eventually with a residual two-body interaction) depending on a deformation parameter q, which in turn is a given function of t. In principle, we should determine self-consistently both the shape of the potential and the dynamical behavior of the system, namely the function $q(t)$. This is not done, however, and we will not need the explicit form of this time dependence for the derivation of a collective Hamiltonian. We will only assume that the motion is slow.

At each time t, that is, for each deformation q, we assume the eigenstates $|k\rangle$ of the Hamiltonian, the so-called *adiabatic basis* [No 77]:

$$H|k\rangle = E_k|k\rangle. \tag{12.138}$$

The eigenfunctions $|k\rangle$, as well as the energies, depend on the parameter q and therefore on the time t.

The exact solution $|\Psi(t)\rangle$ of the Schrödinger equation

$$H(t)|\Psi(t)\rangle = i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle \tag{12.139}$$

is now expanded in this basis:

$$|\Psi(t)\rangle = \sum_k a_k(t)e^{i\varphi_k(t)}|k(t)\rangle. \tag{12.140}$$

For convenience we use time-dependent phase factors

$$\varphi_k(t) = -\frac{1}{\hbar}\int^t E_k(t')\,dt' \tag{12.141}$$

in this expansion, which essentially takes care of the trivial oscillating phase of the adiabatic states $|k\rangle$. Inserting the ansatz (12.140) in the Schrödinger equation (12.139) and multiplying with the vectors $\langle l|$ we obtain a coupled set of differential equations for the coefficients a_k,

$$\begin{aligned}\dot{a}_l &= -\sum_k \langle l|\frac{\partial}{\partial t}|k\rangle e^{i\varphi_{kl}}a_k \\ &= -\dot{q}\sum_k \langle l|\frac{\partial}{\partial q}|k\rangle e^{i\varphi_{kl}}a_k,\end{aligned} \tag{12.142}$$

with $\varphi_{kl} = \varphi_k - \varphi_l$.

This equation is still exact. It is now investigated in the *adiabatic limit*, that is, for small velocities $\dot{q}$. We assume that the system for $t=0$ in the state $|0\rangle$ (for instance a Slater determinant, where all the levels below the

fermi surface are filled) is:

$$a_k(t=0)=\delta_{k0}. \tag{12.143}$$

Since the time derivative of a_k is proportional to $\dot{q}$, we expect for all times $a_k(t)$ to be small for $k\neq 0$ and a_0 to be close to unity, that is, the system always stays close to the adiabatic ground state. There are only virtual excitations to higher states, which are proportional to the velocity and produce an inertia.

In this approximation, for $l\neq 0$ we obtain

$$\dot{a}_l=-\dot{q}\langle l|\frac{\partial}{\partial q}|0\rangle e^{i\varphi_{0l}}. \tag{12.144}$$

Under the *additional assumption* that $\dot{q}$ *and* the matrix element $\langle l|\partial/\partial q|0\rangle$ have only a relatively small time dependence compared to the oscillation phases φ_{0l}, we can integrate (12.144) and for $l\neq 0$ obtain

$$a_l=\frac{i\hbar\dot{q}}{(E_l-E_0)}\langle l|\frac{\partial}{\partial q}|0\rangle e^{i\varphi_{0l}}. \tag{12.145}$$

We have thus calculated the velocity-dependent terms in the wave function (12.140) in first-order perturbation theory and are now able to express the total energy as a function of the "collective coordinate" q and the corresponding velocity $\dot{q}$

$$\begin{aligned}E(q,\dot{q})&=\langle\Psi(t)|H|\Psi(t)\rangle=E_0+\sum_{k\neq 0}(E_k-E_0)|a_k|^2\\&=\frac{1}{2}M(q)\dot{q}^2+V(q),\end{aligned} \tag{12.146}$$

with the potential energy

$$V(q)=E_0(q)=\langle 0|H|0\rangle_q$$

and the inertia parameter, the well known *cranking formula*,

$$M(q)=2\hbar^2\sum_{k\neq 0}\frac{|\langle k|\partial/\partial q|0\rangle|^2}{E_k-E_0}. \tag{12.147}$$

Using the fact that the many-body functions $|k\rangle$ are eigenfunctions of the Hamiltonian $H(q)$, we can write this formula in a slightly different way:

$$M(q)=2\hbar^2\sum_{k\neq 0}\frac{|\langle k|[H,(\partial/\partial q)]|0\rangle|^2}{(E_k-E_0)^3}=2\hbar^2\sum_{k\neq 0}\frac{|\langle k|\partial H/\partial q|0\rangle|^2}{(E_k-E_0)^3}. \tag{12.148}$$

For a simple Nilsson Hamiltonian $H=H_0-q\cdot Q$ we thus obtain

$$M(q)=2\hbar^2\sum_{k\neq 0}\frac{|\langle k|Q|0\rangle|^2}{(E_k-E_0)^3}. \tag{12.149}$$

In this form, the cranking formula has been widely used in the literature, mainly in connection with Nilsson–Strutinsky-type calculations (see Sec. 2.9) of the energy surfaces for the fission process [SSW 69, Pa 75b, Po 76, RLM 76, Ba 78, Po 78].

In the derivation of this formula we made the assumption that the matrix elements $\langle k|\partial/\partial q|0\rangle$ change very slowly in time. This is certainly not true in the region of *level crossings* [St 77] or pseudo-crossings, as in Fig. 2.22. In Section 2.8.4 we discussed the fact that in such regions the characteristic properties of the wave functions are exchanged. If this exchange takes place in a small q-region, we expect large values for the matrix elements $\langle k|\partial/\partial q|0\rangle$ and the derivation of the cranking formula may no longer be justified.

In fact, it can happen that the system does not stay in its adiabatic ground state when it moves over such a crossing. This can be most easily investigated at an *isolated crossing* of two levels ϵ_1 and ϵ_2 with an interaction matrix element V_{12} (the distance of closest approach between the levels is then $2V_{12}$). In this case, Eq. (12.142) reduces to a system of two coupled differential equations. For a constant velocity $\dot{q}$ and a linear dependence of the levels $\epsilon_{1,2}$ on q, this can be solved analytically. For the probability $P=|a_2(\infty)|^2$, that the system jumps at the pseudo-crossing into the upper level, we obtain the *Landau–Zener formula* [La 32, Ze 32],

$$P=\exp\left(-\frac{2\pi V_{12}^2}{\hbar|\dot{q}||(d/dq)(\epsilon_1-\epsilon_2)|}\right). \tag{12.150}$$

This probability becomes large for increasing velocities, but also for an increasing difference in the slope of the two levels, because in such cases the interaction region where the properties of the two levels are exchanged is rather short. In the formula (12.150), which is an exact formula for this model, we recognize that the function P shows a nonanalytic behavior for small velocities $\dot{q}$ [Sch 77a]. A solution of Eq. (12.142) by an expansion in $\dot{q}$, which gives the cranking formula (12.147) in lowest order, therefore probably corresponds to an asymptotic expansion.

Up to now it is a somewhat open question as to in which situations the adiabatic assumption is justified. Numerical calculations for a fixed path starting at the saddle point of the fission process have shown that a very large part of the wave function is excited due to pseudo-level crossings [SW 75a+b, LPP 75, LPF 77, SW 78]. It could be, however, that in a self-consistent calculation as discussed in Section 12.3.4 this picture is changed very much. The conclusion will certainly depend on the excitation energy of the system.